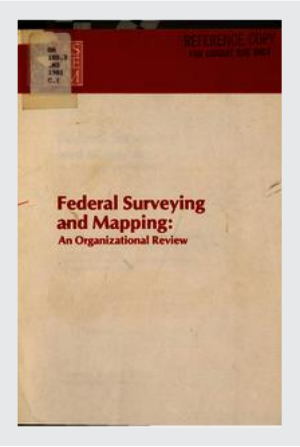
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Federal Surveying and Mapping:
An Organizational Review

Panel to Review the Report of the Federal Mapping Task Force
on Mapping, Charting, Geodesy and Surveying, July 1973
Committee on Geodesy
Assembly of Mathematical and Physical Sciences
National Research Council

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NOTICE: The project that is the subject of this report was approved by the Governing Board of the National Research Council, whose members are drawn from the Councils of the National Academy of Sciences, the National Academy of Engineering, and the Institute of Medicine. The members of the committee responsible for the report were chosen for their special competences and with regard for appropriate balance.

This report has been reviewed by a group other than the authors according to the procedures approved by a Report Review Committee consisting of members of the National Academy of Sciences, the National Academy of Engineering, and the Institute of Medicine.

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Preface

The National Research Council's report *Geodesy: Trends and Prospects* (Committee on Geodesy, 1978, page 5) states: "We recommend that the 1973 Report of the Federal Mapping Task Force on Mapping, Charting, Geodesy and Surveying of the Office of Management and Budget be re-examined and updated. Additional emphasis of this re-examination should be on coordination of the capabilities of federal agencies in earthquake prediction, space techniques, and instrument development."

The objective of that recommendation served as the general directive to this Panel to review the scientific accomplishments in geodesy that might be attributed to that 1973 Task Force report, to identify the scientific and operational surveying and mapping functions that have changed since 1973 because of advances in technology or shifts in national priorities, to identify those activities recommended in 1973 toward which little, if any, progress has been made, and, in so far as possible, to determine the reasons for the lack of response. In addition, the Panel considered it beneficial to identify some of the problems closely associated with surveying, mapping, and geodesy but not considered in the 1973 study and to give special consideration to the 1979 proposal of the President of the United States for the formation of a Department of Natural Resources and to the relevance of such a department to the topics considered in this study.

An examination of earlier studies of federal surveying and mapping activities indicated that, though there has been a significant change in technology, the surveying and mapping functions are essentially the same today as they

were at the time of those earlier studies. In addition, the problems that were noted then (see Appendix B) are also evident today. The many efforts to resolve these problems by consolidating federal surveying and mapping programs under a single agency have not been realized. This Panel has not endeavored to develop an organizational structure for a single agency, but a report by the Federal Board of Surveys and Maps, 1920-1922, recommended such a structure (see Appendix B), which, with some additions and deletions due to some functional changes in the last 60 years, might serve as a model for a reorganization of federal surveying and mapping functions.

The membership of the Panel represents the private engineering sector and the academic community as well as government agencies involved in surveying, mapping, and related activities. The members of the Panel were assisted by staff officers of the Office of Management and Budget, whose expressions of interest in the present study provided encouragement to the Panel.

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1
Summary

1.1 A SINGLE AGENCY

Mapping, charting, geodesy, and surveying are fundamental services and support programs that underlie practically every national endeavor, beginning with the exploration, ownership, and development of the lands, seas, and airspace and their resources. Meeting the increasing requirements for more accurate and up-to-date maps, charts, surveys, and land-related information is critical for the various uses of resources and for the national defense. Collectively such information contributes to the development of a multipurpose cadastre. On a broad national level, attention is focused on future requirements for natural resources and the long-range objectives for the development, utilization, and conservation of these resources. The coordination of mapping, charting, geodesy, surveying, and multipurpose cadastre activities plays a key role in the support of the programs to meet these requirements.

The present situation with respect to the proliferation of surveying, mapping, and related activities among the 39 federal agencies involved is not much different than it was in fiscal year 1972, the year used as the base for the 1973 Federal Mapping Task Force (FMTF) report. The number of agencies remains the same, but there have been significant advances in technology. Some of the specific programs recommended in the 1973 report have been initiated, and there are examples of more effective coordination between agencies. More state and local governments are seeking the technical guidance and cooperative assistance of federal agencies. There is greater emphasis on products and services with special concern about users' needs. Individuals in

the private sector are utilizing the innovations in instrumental and computer technology. There are many positive responses resulting from the FMTF report, yet there has been no definitive action on the key recommendation made in 1973—*that there be a single civilian agency for these activities.*

This Panel strongly and fully endorses the 1973 recommendation for the establishment of a single agency for civilian mapping, charting, geodesy, and surveying and would add the multipurpose cadastre to this list. The 1973 Task Force urged "immediate adoption." We also urge that prompt consideration be given to the formation of a Federal Survey Administration (FSA); however, this Panel prefers the designation Federal Surveying and Mapping Administration to identify more clearly the functions of the organization and to eliminate any ambiguity due to use of the term "survey." A single, centralized agency would provide the impetus and visibility that are required for the fulfillment of national goals and could execute the programs more effectively, efficiently, and economically. Furthermore, we stress that the agency status within a department be independent of other organizational elements whose primary interests lie in other directions.

1.2 ACTIONS RESULTING FROM 1973 RECOMMENDATIONS

The 1973 FMTF report contains an analysis of personnel requirements and budgetary resources available to each agency for fiscal year 1972. This Panel has not attempted to update all aspects of that report. However, there are summary statements relative to the present level of support for the principal surveying and mapping agencies (see Appendix A). Despite the impacts from the effects of increasing inflation and from the restrictions imposed by administratively controlled personnel ceilings, significant progress has been made toward achieving specific goals recommended in the FMTF report.

Much of the budgetary support provided for the following programs or initiatives can be identified as resulting from the recommendations and the overall impact of the 1973 study:

- The new adjustment of the North American Horizontal Control Networks,
- The new (with more than 100,000 km of new leveling) Vertical Control Network for North America,
- The expansion of the Geodetic Data Base to include lower-order surveys,
- The automation of hydrographic surveying and nautical charting,
- The greater use of orthophotographic techniques for mapping,
- The National Cartographic Information Center,
- The Cartographic Digital Data Base, and
- The use of geodetic methods for Cadastral Surveying in Alaska.

1.3 ROLE OF STATE AND LOCAL GOVERNMENTS

This Panel believes that state and local governments should assume greater responsibility for those surveying and mapping functions that relate to regional and local programs. However, the federal, state, and local roles must be completely integrated. The necessity for this coordination is documented in the report *Need for a Multipurpose Cadastre* (Committee on Geodesy, 1980). The following paragraphs from that report (page 102) indicate the overlapping responsibilities of federal, state, and local governments.

> The major obstacles in the development of a multipurpose cadastre are the organizational and institutional requirements. Reorganization and improved quality control for existing governmental functions will be required. Each of the components of the cadastral system already exists somewhere within our existing governmental structure. Many of the required data are being generated at the local level, and in most cases the users are the individual citizens and the local government officials and planning organizations. Each state has organizations that prepare maps and collect data that would be an integral part of a multipurpose cadastre. At present, much of this effort is directed toward a single purpose and cannot be utilized for other programs. The federal agencies conduct surveys, prepare maps, and collect information that would be required for the development of a multipurpose cadastre . . . .
>
> The development of a multipurpose cadastre will require coordination between various levels of government—local, state, and federal, as well as that of parallel organizations at each level.

1.4 ROLE OF THE DEFENSE MAPPING AGENCY

In 1972 the major groups within the Department of Defense that were involved in mapping, charting, and geodesy activities were combined into a single agency—the Defense Mapping Agency (DMA). As a result of this consolidation, it has been possible to establish more weil-defined national priorities and develop more effective procedures for achieving them. The DMA is a good example of coordination and administrative direction. Chapter 4 of this report contains a discussion of the cooperative roles of the military and civilian agencies in support of total national goals. The benefits to be derived from a consolidation of the civilian mapping, charting, geodesy, surveying, and cadastral agencies, in a manner similar to the consolidation that resulted in DMA, are considered.

1.5 PRINCIPAL RECOMMENDATIONS

The primary recommendation resulting from this National Research Council review is that *there be a single agency, a Federal Surveying and Mapping Administration, for civilian mapping, charting, geodesy, surveying, and the multipurpose cadastre.*

The proposed Administration should be an independent agency but could be placed within an existing Department that has some of these functions as part of its mission. If in a Department, the Administration should be independent of other organizational elements whose primary interests are not mapping, charting, geodesy, surveying, and the multipurpose cadastre.

Many of the cooperative geodetic surveys supported, totally or in part, by state and local agencies contribute to the strength of the national networks and will be included in the new adjustments of the horizontal control and the vertical control networks. In addition, there are state and local agency requirements for surveys that cannot be served by the federal agencies, yet the results of such surveys, when properly made and monumented, should be incorporated into the national data base. Similar situations exist with respect to cooperative state and local mapping projects.

We recommend that the mapping, charting, geodesy, surveying, and cadastral agencies of the federal government continue to sponsor cooperative programs, with state and local governments providing sufficient guidance to ensure conformance to national specifications and standards and thus to the development of a fully integrated national information system.

Pending the establishment of a Federal Surveying and Mapping Administration, the existing liaison offices in the various states for the Geological Survey, Bureau of Land Management, Federal Highway Administration, Corps of Engineers, and National Geodetic Survey (a component of the National Ocean Survey) should be utilized for these cooperative programs.

The progress toward the new adjustment of the horizontal control network in North America and the plans being developed for a similar adjustment of the vertical control are described in this report. All the observational data required for these adjustments, as well as the final adjusted data, will be in a machine-readable format. This data base, managed by the National Geodetic Survey, is accessible to those making the adjustments and to other groups, federal and state, involved in cooperative programs. However, at present, there is no provision for the private sector to use this facility.

The Geological Survey is endeavoring to respond to the 1973 Task Force recommendation for the development of a cartographic data base and is seeking the essential budgetary support. The Panel recognizes the magnitude of the task and the unique relationship to the geodetic data base.

We recommend that the geodetic and cartographic data bases be adequately supported, be readily accessible to all users, and, even though serving different interests and needs, be made integral parts of a national mapping, charting, geodesy, surveying, and multipurpose cadastre information system.

In the section of the 1973 FMTF report dealing with nautical charting and hydrographic surveying, emphasis was placed on the full set of navigational products—conventional and special-purpose nautical charts, tide and current

tables, timely notices to mariners, and other special aids to navigation. The requirements for high standards of accuracy to ensure maximum safety were stressed. The Task Force recommended that greater use be made of automated techniques for the acquisition of hydrographic survey data and the compilation of nautical charts. The National Ocean Survey is developing a totally automated system for data acquisition through chart production to distribution. The survey information, in digital format, is entered into the National Charting System Data Bank. *We recommend that the program for the marine data base be adequately supported and that the data base also be made an integral part of a national mapping, charting, geodesy, surveying, and multipurpose cadastre information system.*

The Cadastral Survey Division of the Bureau of Land Management, in response to the Task Force recommendations for greater coordination with mapping and geodetic programs, particularly in Alaska, has identified the tasks, has acquired the most modern instrumentation, and has initiated intensive field programs. However, when considering the overall needs for mapping, geodetic, and cadastral information, the Panel believes that more effective coordination should be achieved. The report *Need for a Multipurpose Cadastre* (Committee on Geodesy, 1980) outlines the benefits to be derived from such a cadastre. Because of the interface with surveying and mapping programs, the development of standards for a multipurpose cadastre and the lead role for coordination among local, state, and federal agencies should be among the responsibilities of a Federal Surveying and Mapping Administration.

We recommend the recognition and support of a national strategy for the multipurpose cadastre within the proposed Federal Surveying and Mapping Administration and the full coordination of the multipurpose cadastre with the surveying and mapping programs.

1.6 CONCLUSION

The Panel commends the 1972-1973 Federal Mapping Task Force for the thoroughness of its study. Significant progress has been made on a large number of recommendations contained in the 1973 report. However, this Panel, after identifying the problems affecting the achievement of important recommendations in that report, believes that many of the difficulties could be eliminated through the consolidated planning and unified program direction from a single agency. Rapidly increasing national requirements for surveying and mapping data and services and the expanding technological growth lead to our conviction that this step is even more important today than it was in 1973. *We urge that prompt consideration be given to the primary recommendation, the formation of a Federal Surveying and Mapping Administration, by the Executive Office of the President.*

2
Federal Mapping Task Force Report, 1973

The principal civilian surveying and mapping agencies—the Geological Survey, National Ocean Survey, National Aeronautics and Space Administration, and Bureau of Land Management—were requested by the Panel to report on the following four items:

1. Progress on the recommendations made in the Federal Mapping Task Force (FMTF) report,
2. Areas in which lack of progress is particularly evident,
3. Areas of interest missing from the FMTF report, and
4. Sections of the FMTF report that should be revised (or updated).

The responses have been used in preparing this chapter. Through informal discussions with representatives from each agency, it was clearly established that, though the principal recommendation in the FMTF report has not been achieved, the report has been useful in support of program initiatives and the associated requests for resources. The progress made by the agencies in response to specific recommendations in the FMTF report confirms this evaluation.

When commenting on the difficulties encountered in responding to the 1973 Task Force recommendations, none of the agencies addressed the principal recommendation, that is, the formation of a single civilian agency. In each of the four major civilian agencies, the surveying and mapping activities are not the primary concerns of the parent organization. Thus, this Panel concludes that the agencies consider this recommendation for a single agency to

be beyond the scope of their primary responsibility. The resolution of such a recommendation involves high-level executive direction more than technical direction. The solution should be achieved by the Executive Branch of the government. We believe that the combined magnitude, importance, and complexity of the existing services as spread through the various agencies merits greater attention and more effective coordination by the Office of Management and Budget. Such action could be the first step toward achieving the goal of a strong central civilian agency. The benefits derived from the consolidation of the mapping, charting, and geodetic activities within the Department of Defense in 1972 to form the Defense Mapping Agency serve as justification for the support of this principal recommendation (see Chapter 4).

2.1 SUMMARY OF THE 1973 REPORT

The Federal Mapping Task Force was established by the Office of Management and Budget in 1971 to study civilian mapping, charting, geodesy, and surveying requirements, operations, products, and methods. This activity followed shortly after the consolidation of Department of Defense surveying and mapping activities into the Defense Mapping Agency. It has been nearly 40 years since the last major study of federal activities in the mapping and surveying area. At that time, the study found a disturbing proliferation and duplication of activities among many agencies. (Previous studies are listed in Appendix B.) Since the time of the last study, this problem has been compounded and a new generation of problems has created additional pressures with the resultant increase in duplication of activity and, perhaps more importantly, further lag in services relating to coverage and timeliness.

The 1973 Task Force was asked by the Office of Management and Budget to

- Determine federal expenditures for mapping, charting, geodesy, and surveying activities and to comment on how best to use these resources in accomplishing overall objectives.
- Identify requirements and comment on their validity as well as the adequacy of programs to meet them.
- Investigate management and program improvements to increase the responsiveness to user needs as well as to improve efficiency and economy of operations.
- Identify and recommend the uses of advanced technology in civilian mapping, charting, and geodetic activities.
- Formulate a comprehensive national program efficiently and effectively to meet the present and future needs and recommend an organizational structure best suited for conducting these operations.

The 1973 Task Force found that 39 federal agencies spent about $305 million for domestic mapping, charting, geodesy, and surveying activities in fiscal year 1972. Less than one third of this effort, approximately $90 million, was spent by the two primary civilian mapping agencies, the Geological Survey and the National Ocean Survey. About $50 million of the total was found to be funded by agencies such as the Department of Housing and Urban Development, the Federal Highway Adminiistration, and the National Science Foundation in the form of grants and contracts. The remaining approximately $165 million was spent by some 34 agencies—either in-house or through contract for a variety of activities. The activities were dispersed among 7 departments and 11 independent agencies. The operating elements were scattered across the nation, in 130 offices and many smaller installations.

On the question of how well the community was doing the job of putting together a national program, the Task Force concluded that there was no one in charge and that proliferation continues. There was some coordination of activities, but generally there was overlap, agency competition, and some fairly serious shortcomings with respect to meeting important national needs. It was concluded that Office of Management and Budget Circular No. A-16 (see Appendix C), issued to overcome some of these problems, had only been partially successful. The lead agencies, the Geological Survey and the National Ocean Survey, were not given a clear mandate to search for and identify duplication nor sufficient authority to influence other agencies' programs. The Task Force found that the research and development effort in the civilian agencies was small and fragmented with insufficient resources to address current and pressing problems. They further noted that advanced technology, developed by the Department of Defense, had not been fully exploited and that significant advantages could be realized if this step were taken. With respect to user requirements, the Task Force was concerned that no mechanism existed in the civilian community for the orderly review, identification, and validation of user needs. They identified several areas of deficiency with respect to coverage, availability of products and data, adequacy of existing products, and programs. Some of the significant program deficiencies had to do with the need for a national urban surveying and mapping program, the need to create a central data and information service, and, for the future, the need to understand and deal with the matter of digital cartography.

Many program recommendations were offered to enhance the efficiency of operations to better meet national needs. Significant related recommendations included the following:

- The need to adjust and upgrade the National Horizontal Geodetic Net.
- The opportunity to build a truly multipurpose network in Alaska by coordinating cadastral, geodetic, and mapping surveys.

- The need to increase vertical control activities significantly to meet currently identified and future requirements.
- The need to coordinate facility, cadastral, geodetic, and mapping control surveys with the objective of building a truly national multipurpose control system.
- The requirement for the establishment of a national urban mapping and surveying program to bring together many uncoordinated and sometimes inadequate activities.
- The need to increase the availability and to upgrade the currency, coverage, and content of topographic maps and to build more flexibility into the national mapping program.
- The need to coordinate several mapping programs of various agencies and to develop a national multipurpose intermediate-scale product.

In the technical area, the Task Force made several recommendations in each of the functional areas of surveying and mapping. Significant related recommendations included:

- The need to use Department of Defense advanced technology in the civilian communities and the subsequent requirement for a central civilian focal point to assure that the technology was made available.
- That data from the Department of Defense be declassified and made available conveniently to the civilian community—especially bathymetric and gravity information.
- The creation of a national photography data base and the establishment of a mechanism to assemble, review, coordinate, and validate civilian agency photo requirements.
- The establishment of a common reference system (standards and specifications for using and displaying spatial data and coordinating the development of information systems related to this area).
- Consolidation of research and development activities within the civilian community and the establishment of a formal agreement with the Defense Mapping Agency whereby joint Department of Defense–civilian research and development projects can be carried out.

The Task Force made many organizational, technical, and program-related recommendations with regard to the effective, efficient, and economical operation of civilian mapping, charting, geodesy, and surveying activities, but the principal recommendation had to do with the creation of a central civilian surveying and mapping agency. As proposed, the agency would have well-defined authority to manage and coordinate the activities of the entire civilian federal surveying and mapping community. Basic national programs and

activities that are potentially systematic and multipurpose would be transferred intact to form the nucleus of a new agency. The Task Force believed that the agency should be of size and scope commensurate with the magnitude, importance, and complexity of the job and must be given sufficient authority to coordinate those activities outside national programs. Such an organization could give needed leadership and assure flexibility to meet current needs as well as new needs as they are identified.

2.2 PROGRESS TOWARD ACHIEVING FEDERAL MAPPING TASK FORCE RECOMMENDATIONS

2.2.1 Horizontal Control

The first section of the 1973 report, Land Surveys, includes recommendations directed toward the national geodetic networks. The primary emphases and thrusts point toward the need for a new horizontal control network that is properly referenced and supported by both geodetic satellite observations and a precise transcontinental electronically measured traverse network. To support this effort further and to maximize the benefits, recommendations were made for the expansion of the data base to include "selective control of less than second-order accuracy," the continuance of cooperative programs when practical, and the transfer from other agencies of "funds which are now expended . . . for control surveys of unknown quality and reusability."

The National Geodetic Survey responded by initiating, in fiscal year 1974, a project for the new adjustment that involved geodetic agencies of Canada, Mexico, and Denmark (for the network on Greenland). The time for completion had to be extended from the recommended 5 years to 8 years because of a stretching out of resources, which will delay completion to 1983. The resources for the U.S. portion of this continental adjustment are provided primarily through redirected program funds augmented by a program increase of $750,000 per year. Assistance is being provided by the Geological Survey for the development of the data base, and there has not been any other infusion of new or reprogrammed funds.

The project has been enhanced by additional geodetic Doppler satellite positioning and horizontal control surveys performed in cooperation with the Department of Defense, the Corps of Engineers, the Bureau of Land Management, and state and local agencies. However, there is concern about remaining on schedule owing to a change of the primary computer system, with associated software conversion problems, and to administrative personnel ceiling policies. Nevertheless, we believe that the recommendation in the report *Geodesy: Trends and Prospects* (Committee on Geodesy, 1978), which en-

couraged the early completion of the new adjustment is still valid, and we concur with that recommendation.

Substantial progress has been made toward the expansion of the geodetic horizontal control data base to include third-order control (Federal Geodetic Control Committee, 1974; 1980).

We recommend that the horizontal reference data base be completed as soon as practical, that the data base be made readily accessible to all national users, and that the data base be made an integral part of a consolidated national mapping, charting, geodesy, surveying, and multipurpose cadastre information system.

Furthermore, we recommend that all surveys of third order and higher be monumented and described for inclusion in the national geodetic data base.

2.2.2 Vertical Control

The 1973 Task Force recommended "doubling the national vertical control program" and suggested that some of the additional funds be secured "by better coordination of surveys that are now single purpose ($400,000) and by reallocating $800,000 from other federal land-survey programs." The Task Force, recognizing the geodynamic aspects of the problems related to maintaining an accurate vertical control network, proposed that primary lines be releveled on a 10- to 30-year cycle, depending on area needs. They found that "about half of the bench marks established more than 30 years ago are lost"; that is, have been destroyed by man or nature.

In 1975, the National Geodetic Survey (NGS), in response to the recommendations of the 1973 Task Force and in recognition of the deficiencies in the existing vertical network, proposed a new vertical control network. During the past 5 years there has been a significant increase in the support of this program. Other federal agencies, working through the Federal Geodetic Control Committee, have upgraded surveying techniques to national network accuracy standards. However, no federal land survey program funding has been reallocated as suggested in the 1973 recommendation.

In fiscal year 1978 the vertical control program of NGS was increased by $2.6 million per year, resulting in a total of approximately $3.3 million per year for leveling. In 1980, the program was extended from a 7-year to a 9-year effort, with an estimated completion date of 1987, in order to achieve compatability with the schedules of the other North American countries and to account for the increased cost resulting from inflation. The program calls for releveling selected lines, 110,000 kilometers, of the primary network and a new adjustment of the total network of approximately 500,000 benchmarks. This international geodetic project involves geodesists from Canada, Mexico, Central America, Denmark, and the United States.

The new leveling must give consideration to the various tidal datums, the International Great Lakes Water Levels Datum, the seaward boundary problems, the ocean geoid, and other measurements related to the surface of the sea. In addition, the new leveling will aid in monitoring regions of crustal movement, such as areas of subsidence caused by the withdrawal of water, petroleum, gas, or other materials, and areas of postglacial rebound or tectonic uplift due to geothermal causes.

This Panel concurs with earlier recommendations that the National Geodetic Survey continue with its program for a new vertical control network.

2.2.3 Gravity

Gravity in the Task Force report was included under Geophysical/Geological Surveys and Mapping, and the recommendations dealt with data classification, marine geophysical survey and mapping programs, and the planning and coordination of federal activities in marine science and engineering. Satisfactory progress has been made and continues, in response to the FMTF, on the declassification and release of gravity data by the Department of Defense to the civilian community. The Geological Survey and the National Oceanic and Atmospheric Administration have used classified gravity data for various studies and have been authorized to publish unclassified contour maps of these data at scales of 1:250,000 and smaller.

The Defense Mapping Agency, the Geological Survey, and the National Geodetic Survey are cooperating in the preparation of gravity anomaly maps for the United States and North America. These maps will be based on data holdings in the gravity libraries of the federal agencies, in the archives of Mexican and Canadian authorities, and in the private sector. This effort is being coordinated through the Gravity Anomaly Map Committee of the Society of Exploration Geophysicists.

2.2.4 Aerotriangulation

The 1973 Task Force recognized the potential of "fully advanced aerotriangulation techniques" for a multipurpose survey. The Geological Survey's major effort to develop such a capability was for mapping in Alaska. Because of the urgency for the National Petroleum Reserve–Alaska, mapping time was not available for acquiring additional or reliable map control in the field nor for taking new photography. The additional control points needed for bridging Air Force mapping photography were established by aerotriangulation using the DMA multistation analytical triangulation (MUSAT) program to adjust observations and to analyze errors in the ground control. The points established from this adjustment were used with the GIANT (a version of MUSAT)

aerotriangulation computer program to establish pass points with the Air Force photography and thus establish control for about 55,000 square kilometers.

The photogrammetric group within the National Ocean Survey has perfected the use of analytical aerotriangulation for establishing survey control. After the completion of a major test project near Casa Grande, New Mexico, the group undertook a project for Ada County, Idaho, which provided coordinates meeting geodetic standards for more than 300 control points. All Bureau of Land Management section corners in the area are included. The project is multipurpose–geodetic, cadastral, and large-scale mapping photography.

2.2.5 Mapping

The formation of the National Mapping Division was implemented by the Geological Survey in 1980, which resulted in the consolidation of the former Topographic Division, the former Publications Division, and the former Geography Program of the Land Information and Analysis Office. The new National Mapping Division (see Appendix A.1) comprises four regional Mapping Centers; a Printing and Distribution Center with regional staffing and activities; and three staff support offices of Plans and Operations, Research, and Information and Data Services. This reorganization represents a major undertaking by the Geological Survey to consolidate its internal surveying and mapping programs and activities.

The 1973 Task Force recommended that the Geological Survey dramatically expand the use of advanced technology for interim revision operations in the National Mapping Program. The Geological Survey has changed the term "interim revision" to "photorevision," which more accurately defines the type of revision used to update the existing maps. Considerable effort has been expended to provide the equipment and advanced techniques needed. A greater effort will be required. The National Mapping Program is making extensive use of special source materials to expedite the photorevision process. An additional facility has been established at the Mid-Continent Mapping Center to handle special source materials, and this will expand the map revision capability of the organization.

The concept of a National Mapping Program, consisting of various interrelated components as recommended in the FMTF report, has been embraced by the Geological Survey. Department of Interior directives have been issued defining the program and setting out operational procedures within the Department. The National Cartographic Information Center (NCIC) was established in 1974. Fourteen federal agencies have entered formal agreements to provide the NCIC with information regarding their data, and 11 states have become NCIC affiliates.

The Geological Survey is now producing orthophotomaps for about 575,000 square kilometers per year (compared with about 150,000 square kilometers of the new 1:24,000-scale topographic maps). These orthophotomaps have been prepared for those areas yet unmapped at 1:24,000 scale and in areas identified to support programs of other federal agencies, such as the Bureau of Indian Affairs, the Bureau of Land Management, and the Soil Conservation Service. The objective to provide orthophotoquads of all unmapped areas of the United States has essentially been accomplished, except in a few scattered areas.

2.2.6 Cartographic Data Base

The recommendation for the development of a cartographic data base has had strong support within the Department of the Interior. In response to an initial request for fiscal year 1979, the Geological Survey's budget was increased by $2 million per year for digital data production. The initial phases of this program included the acquisition of special equipment and the development of new techniques. Digital elevation models were produced as part of the orthophoto program, and other digital data were to be collected during the new mapping production process.

The demands by both federal and state agencies for digital cartographic information systems are increasing rapidly. When adequately funded, the Geological Survey intends, as a long-term goal, to provide cartographic data in digital form for all of the United States, to maintain a national digital data base, and to offer digital service to other agencies.

In support of further developing a national multipurpose Digital Cartographic Data Base (DCDB), studies and assessments continue to be undertaken that focus on this issue. The three most recent investigations are (1) an Office of Management and Budget questionnaire to federal agencies inquiring about agency requirements for a DCDB; (2) a Geological Survey cost-benefit study to support an appropriation issue paper for the fiscal year 1982 budget; and (3) a study by the Office of Science and Technology Policy (of the Executive Branch), to assess the need for a centralized DCDB (Donelson, 1980).

The fiscal year 1982 budget includes a request for $6.034 million for the U.S. Department of the Interior for a digital cartography program. The request also stipulates that legislation be proposed to establish a revolving fund that will finance the program. It is expected that amounts charged for digital cartographic products by the Geological Survey will eventually be the primary revenue source for the fund.

The Geological Survey has been requested to operate the digital cartographic program on a sound financial basis. The agency plans to fund a marketing, pricing, and capital investment study under contract to a private consulting group. To coordinate federal digital cartographic data further, an OMB

Circular is being prepared that will assign the responsibility for this coordination to the Geological Survey.

We endorse the effort of the Geological Survey for the development of this digital cartographic data base and strongly recommend that adequate support for this data base be provided.

2.2.7 Nautical and Aeronautical Charting

The FMTF review of the hydrographic surveying activities of the federal agencies primarily recommended increased use of automated techniques for data collection and processing. The Task Force believed that a more efficient and effective schedule for chart production could be achieved and that the investment costs would yield significant savings and net cost avoidance.

The National Ocean Survey (NOS) has made significant progress toward achieving the goal of a fully automated system for nautical charting. Under data acquisition and processing there has been a 30 percent improvement in the efficiency of hydrographic surveying. Twenty-two systems, known as HYDROPLOT, have been installed on National Oceanic and Atmospheric Administration (NOAA) ships and survey launches. HYDROPLOT systems have also been installed on some Corps of Engineer survey launches. Modern computer/plotter and digitizing systems have been installed at both the Atlantic and Pacific Marine Centers of the NOS. The annual output from these centers has doubled. The survey data, in digital format, are entered into the Nautical Charting System Data Bank. The digitizing of all earlier survey records and chart information should be completed by fiscal year 1984. Copies of the hydrographic data on magnetic tape are forwarded to NOAA's Environmental Data Information Service Center in Boulder, Colorado, for distribution to the public.

The system also provides for computerized compilation and production of charts. The essential tide and current information is entered into the system. The scales include 1:5000 for harbor charts, 1:10,000 for harbor and entrance charts, and 1:20,000 for coastal charts. The NOS has projected that by fiscal year 1986 the total expenditures for automation will be $64 million. Without automation, costs would reach $98 million. Hence automation will result in an estimated cost avoidance of $34 million.

The FMTF also made specific recommendations for the increased use of automated techniques in the production of aeronautical charts and supporting data. Initial steps were taken by the aeronautical charting group in the NOS to assess the feasibility and to prepare a program development plan for automation. A contract for the project was awarded to a firm in the private sector late in 1978, but the contractor was unable to maintain established schedules and the contractual arrangement was severed in 1980.

The Office of Aeronautical Charting and Cartography in the NOS has

been authorized to proceed with the development of this project utilizing "in-house" personnel. This Panel endorses this action for it believes that most of the essential hardware and much of the basic software are available. By employing engineering development techniques, the NOS group will be able to assemble a workable system.

The NOS has been successful in implementing an integrated automated distribution system for nautical and aeronautical charts plus the supporting data. The total number of the various charts produced annually by the NOS is approaching 25,000,000. In addition to the primary distribution by the NOS, there are more than 4000 sales agents for these products. In meeting the needs of civilian and military aviation with the rapid advances in aerotechnology, the efficient distribution of these navigational aids is essential. This final phase is a vital part of the total automated system.

2.2.8 Cadastral Surveys

The Cadastral Survey Division of the Bureau of Land Management (BLM) is responsible for the legal cadastral surveys required by all federal agencies. The 1973 Task Force, recognizing the complementary nature of the rectangular land system use for cadastral surveys and the national geodetic horizontal control network, made two recommendations related to the cadastral survey program. They recommended "transferring appropriate cadastral survey functions of BLM (excluding corner search and boundary marking) to" the new Federal Survey Administration. Although this step has not been achieved administratively, progress has been made through mutual understanding and agreements between the various federal agencies involved.

The second recommendation was "implementation . . . of a combined geodetic/cadastral program in Alaska to meet requirements for State selection boundaries and for other cadastral surveys in Alaska, as practicable, and to provide geodetic control for multipurpose use." The BLM has initiated a vigorous program for cadastral surveys in Alaska. The more complete coordination with the geodetic and mapping programs has not been achieved. These problems are discussed in Section 2.3.5.

Progress by the BLM Cadastral Survey Division since the 1973 report includes the acquisition of inertial survey systems and Doppler receivers for satellite surveying and the introduction of these modern geodetic techniques in their survey procedures. They have developed Memoranda of Agreement with the National Geodetic Survey and the Geological Survey to achieve more effective coordination. These are the types of actions identified by the 1973 Task Force that are required for establishing a national cadastral survey program.

2.2.9 Space Techniques

The National Aeronautics and Space Administration (NASA) has been working with the mapping, charting, geodesy, and surveying agencies and has made substantial progress in the development of space technology to support the operations of these agencies. These techniques include applications for monitoring polar motion and crustal movement, for determining the earth's gravity field and the shape of the geoid, and for various types of mapping including the altimetry of the sea surface. NASA, the National Geodetic Survey, and the Geological Survey are working with the Defense Mapping Agency on geodetic applications of the Global Positioning System, which is under development by the Department of Defense. Because of the application of space techniques during the past 20 years, the uncertainties of determining coordinates on the earth's surface have been reduced from tens of meters to a small fraction of a meter. During the same period, the uncertainty of the radius of the earth has been reduced from 100 m to 1 or 2 m.

2.2.10 Data Banks

Much of the progress in surveying and mapping of the United States during the past decade is due to the effective utilization of computer technology and the development of data bases. Each of the basic functions–topographic mapping, marine system (hydrographic surveying and nautical charting), aeronautical charting, and the geodetic reference system–has its unique data bank. The requirements for these data banks have been identified, and the necessary resources for each data bank have been partially programmed. However, when considering the potential benefits to be realized from the systems, there must be greater coordination between the individual data systems managers, leading to a fully integrated national mapping, charting, geodetic, surveying, and cadastral information system. We believe this can best be accomplished within a single civilian federal surveying and mapping agency.

2.3 PROBLEMS TOWARD ACHIEVING FEDERAL MAPPING TASK FORCE RECOMMENDATIONS

Many of the difficulties encountered by the agencies when endeavoring to respond to the recommendations of the 1973 report may be traced to the lack of coordination at the executive level. The federal agencies have established working groups for the purpose of achieving greater coordination within programs. Agreements have been reached on technical aspects, but difficulties remain relative to the availability of resources such as personnel and appropriated funds. The mechanism provided by the Office of Management and Budget

is its Circular No. A-16 (see Appendix C) on coordination of surveying and mapping activities, but it does not include all aspects of surveying and mapping nor does it provide adequate authority for assuring effective coordination.

2.3.1 Single-Purpose Projects

The 1973 Task Force, in identifying users of geodetic information, stated that approximately 50 percent of the requests for horizontal control came from the federal government. The Federal Geodetic Control Committee, functioning under the directives of OMB Circular No. A-16, endeavors to coordinate these requirements for geodetic control, but the National Geodetic Survey has been unable to respond to all the requests for surveys from other federal agencies. The result is that single-purpose surveys, which do not always contribute to the national network, are made by these other agencies, either in-house or by contract. This Panel believes that much of the economic loss resulting from these practices can be eliminated by more specific coordination by a single civilian agency.

A similar situation exists with respect to mapping. Several federal agencies (primarily the Department of Housing and Urban Development, the Environmental Protection Agency, the Corps of Engineers, and the Federal Highway Administration) finance mapping activities to meet specific individual program needs such as flood insurance, community planning and development, utility planning and construction, and highway mass transit. Hundreds of millions of dollars have been expended for these efforts with little coordination between programs, particularly for the assignment of responsibility to provide accurate multipurpose maps that could serve all users.

In agreement with the FMTF recommendation that funds expended by the Department of Housing and Urban Development clients for urban surveys and maps be transferred to a surveying and mapping agency, we further recommend that it be expanded to include all similar funding in other federal agencies and that OMB Circular No. A-16 be revised to provide for the enforcement of this recommendation.

2.3.2 Directive for Coordination

The present OMB Circular No. A-16 provides general guidance for the consideration of surveying and mapping activities of those federal agencies whose primary missions are not those particular functions. The Circular states: "...evaluation of the potential contribution of those activities to a national surveying or mapping program should be made by such agency in consulta-

tion, or pursuant to a general agreement, with the responsible agency . . . ." These guidelines are not being followed by all agencies even though the responsible or lead agency has been designated. This lack of coordination continues to be a major problem.

We recommend that OMB direct that surveys using federal funds be coordinated and be subjected to consideration by a lead agency for inclusion in a national network.

2.3.3 Cooperative Programs with State and Local Agencies

The 1973 study included consideration of the contribution of surveys made by state and local agencies to the national geodetic networks. Frequently such surveys are made under cooperative agreements with a federal agency. During the past 20 years, cooperative surveys have been made in many of the states. The progressive recognition of the need for horizontal and vertical control by state, local, and private agencies has produced an overwhelming number of requests for control survey information. It has not been possible to respond to all of these requests, as the allocation of survey resources has been subject to federal requirements, primarily those related to high-priority programs such as the new national networks of horizontal and vertical control. The potential benefits to be derived from these cooperative programs have not been achieved because of limited resources.

However, we believe that cooperative programs, when properly managed, should be continued. Positive responses to such requests should be premised upon the following:

- Potential of multipurpose use of specific programs;
- Relative population density, land value, and resource potential;
- Relative station benefit–cost ratio; and
- Availability of trained personnel.

We recommend that the federal mapping, charting, geodesy, and surveying organizations:

(a) Continue to sponsor cooperative programs with state and local governments and that the budgetary procedures recognize these programs, and

(b) Maintain sufficient liaison representatives in each state to monitor the surveying and mapping activities in the community, coordinate the activities involving federal funds in order to avoid duplication and waste, and encourage the use of national specifications and standards so that local projects may be incorporated into the national surveying and mapping programs.

2.3.4 Crustal Movement

When the 1973 FMTF considered the time-varying aspects of geodesy, they recommended "that the National Geodetic Program manager be given responsibility to (1) coordinate all federal civil earth-model development activities that contribute to the National Geodetic Networks, and (2) acomplish geodetic measurements–that is, positions, elevations, distances, and changes in distance–to support polar motion and crustal movement investigations or studies."

The studies of plate tectonics and lithospheric deformations have broadened the participation in crustal movement investigations. The National Research Council report *Geodetic Monitoring of Tectonic Deformation–Toward a Strategy* (Committee on Geodesy/Committee on Seismology, in preparation) investigates a range of methods and issues relating to geodynamics, with primary emphasis on global and continental aspects, and recommends programs and procedures for monitoring crustal deformation. The following paragraphs are from the Introduction to that report:

> The United States Geological Survey (USGS) and the National Geodetic Survey (NGS) have been carrying out, for some decades, programs of monitoring the strain over the tectonically active fault systems of the western United States. Monitoring during the 1970's of trilateration networks has been particularly important. The interpretation of crustal motions from triangulation data have also been valuable. In recent years, much effort has been devoted to repeated leveling surveys performed to monitor possible large and horizontally extensive vertical motions in southern California. We express our strong support for continuation and extension of the trilateration network measurements and of some other parts of these programs. We discuss the need for, and suggest improvements in, the application of classical geodetic techniques and discuss additional means of monitoring crustal motions. We also encourage more extensive scrutiny, utilization, and interpretation of the existing data base.
>
> The principal goal is to improve our knowledge of spatial and temporal patterns of strain accumulation. We believe that it is important to cover a relatively large range of distances in which a change in the pattern of deformation might occur. For this reason, the baseline length of suggested measurements ranges from 1 to 100 km and the sampling rate from one sample per 10 years to one sample per second.

The National Geodetic Survey (NGS) is the agency responsible for the horizontal and vertical control networks and the respective data base to meet engineering requirements and actively participates in the scientific programs supporting geodynamics. The engineering and scientific aspects must be unified so they complement, not duplicate, each other. The engineering tasks are of such magnitude that state and local agencies must accomplish some of the measurements and forward observational data to the NGS for analysis and adjustment into the data base. Consideration could be given to the use of federal grant support for such efforts.

The 1973 recommendation relative to NGS management is still valid for survey activities. Interagency working groups have been established for broader geophysical applications. The problems are well defined, the observational techniques are being perfected, and the operational schedules are being established; but the resources required to implement the programs have not been provided to support all the agencies involved adequately. Their planning could be more effective with greater coordination at the executive level.

2.3.5 Cadastral Surveys

The 1973 Federal Mapping Task Force made several suggestions and recommendations directed toward the cadastral surveying activities of the federal government. The primary federal group involved in cadastral surveys is the Division of Cadastral Survey (formerly the Surveyor General, General Land Office) of the Bureau of Land Management (BLM), which has the responsibility for the cadastral survey of public lands. However, there are at least nine other federal agencies engaged in some aspect of cadastral surveying, much of it not involving public lands. The BLM has responded to Task Force suggestions by introducing modern electronic distance-measuring instruments as standard equipment, Doppler receivers for connecting the rectangular land survey system to the national geodetic network, and inertial positioning systems for use in Alaska. This use of modern technology is in sharp contrast to the statement in the 1973 report: "Cadastral surveys are of minimal value to the geodetic surveying community, and positions and areas based on the cadastral network are not directly relatable to the geodetic network."

Nevertheless, this Panel believes that the response has not been as full as it might be and makes the following recommendation:

Recognizing the extensive effort in cadastral surveys in Alaska and the need to expand the data base to include monumentation, documentation, and photoidentification of points to serve the needs of the mapping and geodetic programs, we recommend that the cadastral effort be more fully coordinated with other federal mapping and geodetic agencies. When considering the total needs in Alaska for mapping, geodetic control, and cadastral surveys, this maximum coordination is essential.

The fundamental problem, as seen by this Panel, relates to resources. At present, the BLM is not adequately funded for its program in Alaska, nor are the other federal mapping and geodetic agencies provided the resources to upgrade their level of activity in Alaska to assure the recommended coordination. A central surveying and mapping agency could focus on these problems and could initiate programs to alleviate the situation.

There has been no action on what might be considered the primary recommendation of the 1973 report relative to cadastral surveys, that is

"transferring appropriate cadastral surveying functions of BLM . . . to a new central agency." Such action requires legislation or an executive order.

2.3.6 Digital Data Bases

In Section 2.2.6, reference is made to the progress that the Geological Survey has made in the development of its cartographic digital data base. Also, the geodetic data base is discussed in Section 2.2.1 and the Nautical Charting System Data Bank in Section 2.2.7. There are basic philosophical differences among the cartographic, geodetic, and nautical chart digital data sets that should be noted. The cartographic data base is being formed by digitizing the various elements of the 1:24,000-scale topographic maps. The geodetic data base contains the original observations and the adjusted data for the horizontal and vertical control networks. The nautical chart data bank contains the hydrographic survey information, that is, all soundings referenced to point coordinates, in addition to all the supporting navigational information that may be required for any chart at any scale. However, the whole subject of digital data development and automation of the mapping process merits more attention than it is receiving. Coordination is taking place, but on a "volunteer" basis, so with involvement by several agencies there could be unnecessary duplication of equipment and production effort. Also, the information collected may not be compatible or interchangeable.

Considering that the digital data base program recommended in the Federal Mapping Task Force report falls far short of today's requirements, we recommend the development of a data-management system that incorporates the activities of the mapping, charting, geodesy, surveying, and cadastral agencies and that will lead to an integrated land-information system. We furthermore recommend that a lead agency be designated for this program and that sufficient new funding be made available to accomplish the task.

2.3.7 Research and Development

The Federal Mapping Task Force made recommendations for more effective use of the research and development (R&D) resources of the various agencies. The 1973 report described the R&D programs in mapping, charting, and geodesy (MC&G) of the civilian agencies as "small and dispersed" or "fragmented." The level of funding was not sufficient to "develop the expensive and complicated automated systems necessary to meet growing domestic needs effectively and efficiently." It may be concluded that the Task Force was giving more attention to "development" than to "research." There are basic and applied research programs supporting surveying and mapping that merit equal consideration and may not require the same financial support as

needed for systems development. The combined effort of all civilian groups should be subject to periodic review and evaluation.

Recognizing the efforts by federal civilian agencies to coordinate their research and development programs in mapping, charting, geodesy, surveying, and the multipurpose cadastre through ad hoc *cooperative agreements and technical exchange programs between agencies, we recommend that this coordination be formally established either by designation of lead agencies or by revision of OMB Circular No. A-16 to provide for such coordination as an interim action until the time a single Federal Surveying and Mapping Administration is formed.*

2.4 PROGRAM UPDATE SINCE THE FMTF REPORT

The advances in technology and the shifts in national priorities are such that there are several major activities involving surveying and mapping that need to be identified and included in a review and update of the 1973 FMTF study. These include the scientific emphasis on national and international studies relating to geodynamics, the increasing importance of cadastral surveys, the maintenance of the federal highway system, the shifts of population toward the coastal zones with the associated development, and the federal support for urban renewal and community development.

2.4.1 Geodynamics

The section of the Task Force report dealing with geodetic support of geodynamics is considered inadequate in view of the many important questions that are beyond our present capability to answer, despite the rapid progress achieved within the past decade in understanding the dynamic behavior of the earth. Some selected examples are: How are the tectonic plates moving at present? Is the movement smooth, or is it episodic? Almost the only information to date on contemporary plate movement comes from earthquake focal mechanism estimates, but how do these sudden movements at active plate boundaries relate to the gross movement of the plates? How do the plates deform in response to the driving forces responsible for the movement? How is the strain distributed near active plate margins, and how does the strain change with time? What relationship, if any, is there between variations in polar motion and other geodynamic phenomena such as great earthquakes?

What these questions have in common is that progress in finding answers to them depends at least on our ability to measure geodetically the relative positions of points on the earth's surface and on changes of positions over distances ranging to many thousands of kilometers. For the determination of relative positions of points and their changes over distances shorter than 100

km in determining intraplate deformations, such as measuring motions along fault zones, classical ground-based geodetic surveying methods are useful and important. However, for baselines longer than 100 km there are several reasons why such techniques are impractical. The basic one is that ground surveys must necessarily be made in a series of line-of-sight measurements between points a few kilometers apart, and the accumulation of errors soon brings uncertainties in position above the required levels of a few centimeters. The second reason is that ground surveys are time consuming and expensive. The cost of frequently monitoring horizontal and vertical movements possibly several times per year in even a few tectonically active areas is prohibitive.

To meet the new needs described above, new ground techniques and space methods need to be and are being developed by a number of federal agencies and other organizations. For ground surveys these include single and multiwavelength electronic distance-measuring devices, high-precision gravimeters, and tilt and strain meters. The most important methods involving space technology are laser ranging to the moon and to man-made satellites and microwave radio interferometry. In addition to these positioning systems, other significant contributions are being made for the detailed mapping of the earth's gravity and magnetic fields using both ground and space technology.

The classical astronomical methods of measuring polar motion and earth rotation appear to have an accuracy limit of about 1 m. Theoretical estimates of the change in polar motion due to a large earthquake suggest that this accuracy must be improved to a few centimeters to be able to study possible relationships between these phenomena.

With the development of space techniques, very-long-baseline interferometry (VLBI), Doppler, and laser-ranging satellite measurements provide an opportunity for improving the accuracy by at least an order of magnitude plus the advantages (for VLBI and Doppler) of all-weather systems. Possibly the most notable effort in this regard is the project POLARIS (Polar Motion Analysis by Radio Interferometric Surveying) initiated by the National Geodetic Survey (NGS) in 1977. The plan specifies that NGS would establish and operate a three-station network of VLBI observatories to monitor polar motion and earth rotation on a continuing basis. Two of the stations will be completed by 1981 (Ft. Davis, Texas, and Westford, Massachusetts). The third station, at Richmond, Florida, has suffered delays caused by funding constraints. In view of the national and international significance of the above triangle, *the Panel strongly recommends that adequate support be provided to complete the VLBI network and to enable its operation in a continuous mode.*

The International Astronomical Union and the International Union of Geodesy and Geophysics through its International Association of Geodesy have formed joint working groups for a special period of collaboration on a

new project known as MERIT (Monitoring of Earth-Rotation and the Intercomparison of the Techniques of observations and analysis). Some agencies have had difficulties in obtaining additional resources for these programs. *This Panel emphasizes the need for international cooperation and acceptance of the results for the reduction of astronomical and geodetic measurements to a common reference or origin. We recommend that federal agency programs for acquiring such data and federal participation in international programs such as MERIT be supported.*

Few of the above-described activities had been foreseen and therefore were not included in the recommendations of the 1973 FMTF study.

2.4.2 Cadastre

The section of the Task Force report dealing with federal cadastral surveys (pages 44-55) implies that the surveys are of minimum value because they were accomplished without regard for geodetic accuracies or ties, even though the surveys were conducted in accordance with federal law. The report further states that the surveys of the conterminous 48 states are all revision surveys, implying that the original surveys are now being "corrected"; the latter is not the case. However, the BLM does execute resurveys that are a reestablishment of the original survey. Local survey units of appropriate land-management agencies perform administrative surveys that define land boundaries between existing monuments.

The report recommends that a dense network (20-km interval) of second- or higher-order geodetic control be accomplished to provide for geodetic survey ties to the BLM rectangular surveys so as to provide cadastral data for a land resource management information program. The BLM plans to provide the geographic position of each rectangular survey southeast corner for use as a parcel identifier in a multiple-use land data bank interactive graphics system. Hence, the Public Land Survey System will no longer be independent of the geodetic reference system when the plan is fully implemented.

The 1973 FMTF report did not specifically address the national needs for a multipurpose cadastre. However, the National Research Council report *Need for a Multipurpose Cadastre* (Committee on Geodesy, 1980) enumerates the benefits to the local, state, and federal governments of a national multipurpose cadastre. It emphasizes that the development of a multipurpose cadastre should be "gradual" and that "many of the functions necessary for the land-information system are being performed now." Coordination between federal, state, and local governments with technical direction and guidance from the federal level are the key factors. The Panel that produced that report stated: "It is imperative that the federal effort be coordinated through a single surveying and mapping agency."

2.4.3 Highways

At the time of the 1973 report, the special surveys required for the development of the Interstate Highway System were considered to be local in nature and were classed as "facility surveys." Now, when we consider the total highway system and the surveying and mapping activities associated with its maintenance, this phase of the program becomes national in extent. The liaison role of the Federal Highway Administration (FHWA) becomes important for the coordination of state and local activities with federal functions in surveying and mapping in support of the highway system.

The mission of the FHWA is to assist state highway agencies, through Highway Trust Fund financing, in the satisfaction of their transportation needs while maintaining a degree of uniformity in the development of an integrated system of federal-aid highways, including the 68,000-km Interstate System, and over 1,500,000 km of federal-aid primary and secondary roads and urban streets.

The funding by FHWA of state activities in surveying and mapping is intended only to support the larger activity of highway development, either in individual construction projects or as part of statewide planning. Accordingly, with but few exceptions, FHWA financing of surveying and mapping is dedicated to the accomplishment of state and local goals through transportation improvement.

The administration of federal-aid highway funds is now largely decentralized, and the responsibility for approval and monitoring of programs and projects has been delegated to the field offices in each state. In some states, surveying and mapping are being separated from the Department of Transportation and included in other state departments under a variety of names, such as Department of Environmental Development, Environmental Resources, and Natural Resources.

The FHWA recognizes the routine need for state and local agencies to make surveys and develop maps that satisfy instant project requirements but may have marginal benefits to national goals. Federal funds are often used for this expedient work. The FHWA, serving as the representative for the Department of Transportation on the Federal Geodetic Control Committee, has been an active supporter of the provisions of OMB Circular No. A-16, especially for the strict compliance with the standards and specifications for geodetic surveys.

A spokesman for FHWA states that "any success experienced by FHWA in implementing national highway goals through the federal-aid highway program has been attained through education and support rather than mandate. Accordingly, the same success in surveying and mapping can be attained through close cooperation at the local level, the continuation of cooperative programs, and persistence in educating state, county, and local surveyors and

mappers in the benefits to be derived from quality maps and accurate surveys" (Johnson, 1980).

In Connecticut, the Bureau of Highways in the State Department of Transportation has made excellent use of this federal-aid support program. Because of the favorable ratio of the combined lengths of the interstate, primary, and secondary systems to the geographic area of the state, the Chief of Surveys in the Bureau of Highways reports that they will be able to complete a high-density geodetic control survey for the state from these funds (Sullivan, 1980). The field surveys are being accomplished by a unit of the National Geodetic Survey under a cooperative agreement that provides for full reimbursement.

2.4.4 Coastal Zone

Mapping, charting, surveying, and geodesy in the Coastal Zone have become focal points of concern and are areas of substantial program emphasis since the 1973 FMTF report was prepared. The demand for National Ocean Survey products and services has increased dramatically since the Coastal Zone Management Act of 1972 and related coastal and offshore legislation were passed.

Indices of these demands are the more than 60 million people who now live and work in counties within 80 km of the Gulf of Mexico and the Atlantic Ocean. The national shoreline population is increasing three times faster than growth rates for other areas. Overall, more than 53 percent of the total industrial base of the United States is located within the coastal region.

Outgrowths of the increased resource development, environmental concern, and new legislation since the 1973 FMTF report are demands for the following:

- Tidal datum determinations and information along shore in estuarine and wetland areas;
- Bathymetric and topographic mapping;
- Marine boundary determinations;
- Extended jurisdiction boundary delimitation;
- Geographic positioning at sea;
- Geodetic leveling information and measurements of subsidence and environmental impacts;
- Correlation of estuarine, tidal, and geodetic elevation data;
- Geodetic positioning (i.e., network development) in the Coastal Zone and offshore; and
- Maps and baseline oceanic and geophysical (such as geoid and sea-surface) information for marine resource assessment, monitoring, and prediction.

In the meantime, the National Ocean Survey has implemented new program efforts to provide:

- Marine boundary determinations in cooperation with the states,
- Geodetic survey and services in cooperation with the State of South Carolina; and
- Bathymetric and topographic mapping in cooperation with the Geological Survey and the states.

"Topographic-bathymetric" maps combine the efforts of the Geological Survey and NOS into a single product. The topography delineates the natural and man-made features of the land. Ocean features include bathymetric contours, offshore protraction survey data, oil and gas platforms and pipeline locations, shipping channels, and shoals. The ultimate objective is complete coverage of the United States Coastal Zones at scales of 1:24,000, 1:100,000, and 1:250,000. At the initial stage, the mapping effort is concentrated on the 1:250,000 series, which will be used primarily for planning purposes.

Fulfillment of the new requirements involves additional initiatives in mapping, surveys, and geodesy, including the development and use of new technology, remote-sensing data collections, and measurements. Some examples of expanded or new activities needed are the following:

- A comprehensive coastal hazards program in response to the threat of hurricanes and severe storms along the Atlantic and Gulf Coasts;
- Increased geodetic surveys and services in cooperation with other federal, state, and local agencies;
- A shoreline mapping and monitoring program;
- Measurements of crustal motions and subsidence in coastal areas;
- Expanded fishery obstruction charting;
- Accelerated storm evacuation mapping;
- Expanded ocean-dynamic measurements;
- A concerted funded effort to correlate geodetic, geophysical, and oceanographic data for comprehensive use; and
- Development and implementation of the Global Positioning System, satellite-sensing fine-grain bathymetry, and other mapping, surveying, and oceanographic technology.

The breadth of interests shown in these demands and activities helps to reenforce the recommendation for a Federal Surveying and Mapping Administration.

2.4.5 Urban Development

The 1973 Task Force, when reviewing the activities of the Department of Housing and Urban Development (HUD), recognized that many of the projects supported by HUD had a fundamental requirement for surveys and maps. The 1973 report contains recommendations that ask for greater coordination with the surveying and mapping agencies and suggests the transfer of funds by the Office of Management and Budget to accomplish this objective.

The programs and priorities of HUD have changed somewhat during the past few years. However, the basic philosophy remains the same. HUD's mission is to increase the supply and quality of housing and housing opportunities especially for low- and moderate-income citizens and improve their environment through building new or improving existing communities. Mainly, two methods are used: (1) technical advice and assistance as well as performance standards and (2) financial aid in the form of direct grants or loan guarantees. HUD, as an agency, makes no surveys, draws no plans, and builds no buildings. Until 1975, HUD programs included a number of categorical grants where specific projects were funded, such as the building or extension of a municipal water or sewer system, direct aids to mass transit, the purchase of land for parks and their development for recreation purposes, and the acquisition of a substandard housing area and its clearing and rebuilding with housing and necessary support facilities.

Beginning in 1976, HUD's various categorical grant programs were consolidated into a single block grant program with an annual range of $3 billion to $4 billion. Roughly 75 percent of this must be spent to benefit low- and moderate-income persons; the balance can be used for almost any municipal improvement for which local tax money would be authorized. HUD reports that "the purchase of maps or mapping services is an eligible expenditure of block grant funds."

However, we believe that very little emphasis has been placed on mapping. An examination of the fiscal year 1978 HUD financial report for "Community Planning and Development" shows that out of a total of $56 million, $200,000 was used for "Mapping" while $26.5 million, almost half, was used for "Management." Another $8 million was allocated to "Land Use–Planning and Programming."

HUD's current interest in urban areas includes sociological aspects to as such or even greater extent than the physical infrastructure of housing, utilities, transportation, and public facilities, which are frequently associated with conventional city planning. HUD operates almost exclusively in urban areas and finds that for its societal programs most of the cities are mapped at scales and accuracies that serve the purpose.

At the time of the 1973 study, HUD was actively engaged in an extensive

mapping program in support of the Flood Insurance Program established by Congress in 1968. The ability to establish the 25- and 100-year flood levels was essential to forecast flood risk and to set insurance rates. The mapping was accomplished primarily under contracts with private groups, with limited assistance from federal agencies. The Flood Insurance Program is now operated by the Federal Emergency Management Agency (FEMA). However, this Panel believes that there will always be a need for surveys and maps to support this program and considers the recommendation made by the 1973 Task Force to be valid: "that flood insurance mapping be accomplished by the new agency, and that standards and specifications be developed in accordance with HUD [now FEMA] requirements."

2.4.6 Conclusion

In conclusion we emphasize the potential role of a single agency to respond to shifts in national priorities and advances in technology. With central management, the duplication of effort in long-range planning would be eliminated. The managers of engineering, scientific, and societal domestic programs requiring surveying and mapping information would have a single point of contact. The national needs, programs, and goals would be more effectively identified and achieved.

3
Organization Alternatives

3.1 SINGLE CIVILIAN AGENCY

In 1973 the Federal Mapping Task Force presented several alternatives for the consolidation of the federal activities for mapping, charting, geodesy, and surveying. We have reviewed these various proposals and strongly support the choice they made—a new civilian organization. Therefore, *we, too, recommend the creation of a central civilian mapping, charting, geodesy, surveying, and multipurpose cadastral agency, which we would call the Federal Surveying and Mapping Administration.*

The statement made by the 1973 Task Force in support of this recommendation merits repeating:

> The Federal Survey Administration will bring together into one agency the significant (MC&G) [mapping, charting, and geodesy] programs (and supporting data-collection activities) that are systematic and national in scope and multipurpose in concept. It will be given the capabilities and mandate required to plan, establish priorities, weigh alternatives and priorities, and operate MC&G programs so that they can provide optimal support to national objectives with quicker and better products. Accordingly, it will be structured to respond vigorously and coherently to Federal, state, and local requirements for organized and accurate spatial data, whether presented graphically on conventional map and chart formats or in digital form via tape or printout. By exercising its new capabilities and by taking aggressive community leadership, we expect the Federal Survey Administration to reduce expenditures for overhead and slow down and then stop the uncoordinated growth of MC&G activities and eventually bring about reduction in single-purpose programs.

We believe that the roles of the federal and state governments should be clearly defined and understood in order to achieve these objectives. The key points of these roles are outlined in the following sections.

3.1.1 Role of Federal Agency

Mapping, charting, geodesy, surveying (including legal survey platting), and the mutipurpose cadastre can and should serve a wide circle of activities. If the proposed Federal Surveying and Mapping Administration is to fill its role fully, it should not be closely allied with any one of the benefiting fields, be it forests, minerals, oceans, or another activity or discipline. Close alliance with one of these as a "parent" organization will tend to limit its effectiveness in serving the needs of the others. Thus the mapping, charting, geodesy, surveying, and multipurpose cadastre activities should be placed together in a separate organization, ready to serve all needs equally.

When the strategy of the Executive Office of the President is determined, the proposal should include the essential actions to define the role of the Federal Surveying and Mapping Administration (FSMA) and its authority for coordination with other federal agencies. These actions must also establish the lead-agency role for each of the various functions or services of the new administration.

This Panel concludes that the plan should also include the following:

- Responsibility for the preparation of statements of the national needs for surveying and mapping and the requirements for carrying out the programs.
- A clear expression of intent that, in the formation of an FSMA, authority be provided for the most effective organization of the mapping, charting, geodesy, surveying, and multipurpose cadastre activities, and for the inclusion of all operational resources such as survey equipment, ships, and personnel.
- Recognition of the fact that some elements of mapping, charting, geodesy, surveying, and multipurpose cadastre activities are carried on in several federal agencies that may not be brought into the FSMA, and provision should be made for the review and coordination of these activities by the FSMA.
- Emphasis on the improvements in service that this consolidation could provide; although potential cost savings are significant, the improvements in service would be more important.

A point of concern has been the relationship of the Department of Defense (DOD) hydrographic and oceanographic survey ships, personnel, and asso-

ciated resources to DMA management. Before DMA was formed the Navy had six ships assigned to survey operations. That number has now been reduced to two, and sometimes only one is in operation.

A more serious aspect has been the division of personnel between the surveying and charting operations in the DOD. The personnel engaged in cartographic activities are not getting survey experience nor are the hydrographic and oceanographic survey personnel assigned to the charting program. There are significant benefits to the total program when the professional personnel have this broader experience. In the National Ocean Survey, with 16 ships in operation, the surveying and charting functions are fully integrated. With the development of a highly automated system it is essential that the ships, personnel, and associated equipment not be separated from the charting program.

3.1.2 Role of State and Local Governments

The Federal Mapping Task Force (FMTF) report mentioned the contributions of state and local governments to the national surveying and mapping efforts in only an incidental manner. It recommended the continuation of cooperative programs for the establishment of control survey networks and mapping projects that do not interfere with national priorities. It recommended that the County Highway Maps and the Urban Mapping programs be continued with emphasis on the coordination and monitoring of the programs by the federal agencies that provide most of the funds for the projects.

The federal mapping and surveying organizations have legislative mandates to perform certain activities that provide the public and the governmental agencies with the necessary surveys and maps for the analysis, planning, and execution of programs to benefit the citizens of the United States. The 1973 FMTF report indicated that the federal agencies use of the most popular maps (Geological Survey 1:24,000 topographic quadrangles) was about 25 percent of the total, while the state, local, industry, and public use was about 75 percent; the federal agencies use of horizontal control surveys was 47 percent, while the locally oriented use was 53 percent. The primary federal mapping and surveying organizations have made a great effort to satisfy the local needs for surveying and mapping products by establishing readily available data centers and increasing the number of local outlets for maps.

Even though the recognition of local interest in surveying and mapping products is evident, there has been little effort to integrate the local capabilities for producing the surveys and maps into the national program. At present, many of the local surveying and mapping projects are directed by federal, state, and local agencies that have only one objective–a single-purpose

survey or map for the particular project. There is a need for centralized monitoring of these efforts, to reduce the wasted effort resulting from the diverse local projects. In several instances, the 1973 FMTF report recommended better coordination, maintenance of standards, use of uniform specifications, and monitoring of local programs by federal agencies.

At one time the federal agencies were the only organizations capable of executing large control survey projects or extensive mapping programs. The recent development of electronic distance-measuring instruments, Doppler satellite-survey methods, and inertial positioning systems enables the state and local surveyor to achieve results comparable with those of the federal agency surveyors. The availability of computers, adjustment programs, data-handling systems, and remote access to federal data files and computer programs provide the local agency with the tools to handle most control survey projects. Photogrammetric techniques, data systems, computer plotting, and copying processes provide extensive mapping capabilities at a local level.

Several states and local governments have established control surveying and mapping organizations; most notably the States of North Carolina, Minnesota, Georgia, and Connecticut and Los Angeles County and the City of Chicago. Many of the state highway departments have capable control surveying and topographic mapping organizations. The work of these organizations could be integrated into the national programs at minimal cost and little increase in personnel. With slight modifications in procedures and formatting, much of the state highway department and Corps of Engineers control surveys could be included in the national data files.

Local expertise and availability are particularly valuable in the maintenance of control survey monuments and cadastral monuments. A state or local representative is more accessible than remote federal agencies in most cases where important monuments are about to be destroyed; it is estimated that the cost of perpetuating a survey monument or erecting a reference monument is about 10 percent of the cost of a complete resurvey to establish a comparable monument.

It is recommended that the Federal Surveying and Mapping Administration:

- *Sponsor cooperative programs with state and local governments and that the budgetary procedures recognize these programs and*
- *Maintain liaison representatives in each state to monitor the surveying and mapping activities in the community, to coordinate the activities involving federal funds in order to avoid duplications and waste, to apply national specifications where federal funds are involved, and to encourage the use of national specifications and standards so that local projects may be incorporated into the national surveying and mapping programs.*

3.2 THE PROPOSED DEPARTMENT OF NATURAL RESOURCES (1979)

3.2.1 Brief Description of the Department

In March 1979 President Carter announced that plans would be developed for a new department covering the field of natural resources. It was indicated that it would consist of most of the present Department of the Interior plus the Forest Service from the Department of Agriculture and the National Oceanic and Atmospheric Administration (NOAA) from the Department of Commerce.

In April 1979, a preliminary plan for a Department of Natural Resources (DNR) was issued by the Office of Management and Budget (OMB) to "provide public information and discussion regarding the purpose and composition" of the new department.

By May 1979 specific plans had been developed for the new department. At that time, however, President Carter concluded that other, more critical, legislative matters should be given primary attention and that, consequently, the proposal for creating a DNR would not be submitted to the Congress.

Although action was set aside at that time, the preparatory work that had been done in OMB developed a full justification and description for a new Department of Natural Resources. The following comments are drawn from or are based on the plan developed in OMB.

In planning the new department, primary consideration was given to the basic resources, including, among others, the oceans, land, forests, and minerals. Comparatively little attention was given to surveying, mapping, and charting. Some pertinent comments were made, however, as follows:

- Civilian mapping and charting activities are performed primarily by the Departments of the Interior and Commerce and are further scattered among numerous user agencies. The extensive study on this subject initiated by OMB (Federal Mapping Task Force Report, 1973) is cited.
- Although NOAA is a major component of the Department of Commerce, the most extensive working relationships for NOAA are outside of that department.
- After noting that NOAA programs are highly compatible with a mission of a Department of Natural Resources, comment was made that NOAA and the Department of the Interior now have several competing or duplicate functions. Mapping and charting was cited as an example.

As planned, a Department of Natural Resources would have consisted of five major administrative units: oceanic and atmospheric resources; land and

forest resources; water resources; geology and minerals; and parks, recreation, and wildlife.

It was stated that "the integrity of NOAA functions" would be retained in the new department. To the present NOAA programs would be added the offshore oil and gas leasing and enforcement activities of the Geological Survey and Bureau of Land Management. Obviously, it was planned that NOAA would remain intact and would become the Oceanic and Atmospheric Resources Administration in the new department.

Similarly, it was planned that the Forest Service would become the dominant force in the Land and Forest Resources Administration, which would include most of the present functions of the Bureau of Land Management.

The Geology and Minerals Administration would consist of the Geological Survey, less its Conservation Division. It was stated that the Geological Survey would continue with three major operating divisions: geologic, water resources, and topographic.

Thus the DNR plan did not specify significant organizational changes in any of the surveying, mapping, charting, geodesy, and cadastral activities at the time of formation of the department. However, it is clear that it was anticipated that regrouping and consolidation within these activities would be worked out after DNR was formed. Surveying and mapping activities of the Geological Survey, Forest Service, and Bureau of Land Management and surveying and charting activities of NOAA are cited as areas in which substantial cost savings could be realized, with specific reference to printing and distribution and to more efficient use of expensive equipment. It seems evident that the reorganization planners in OMB concluded that considerable reorganizing of mapping, charting, geodesy, and surveying should be carried out but chose to leave the detailed planning and action to be worked out after the major components were together in the same department.

3.2.2 Status of Proposed Department of Natural Resources (1979)

Insofar as information is available to the Panel, there is no plan for renewed action on a DNR at this time. It seems evident, however, that there was clear belief in OMB that bringing together the identified activities in a new department was justifiable and that, if accomplished, it would be generally beneficial. The plan in OMB to put together a file on the findings, conclusion, and justification developed during the March–May 1979 period suggests a belief that the completed work may be useful in a reactivation of the proposal in the future.

3.2.3 Shortcomings of Proposed Department of Natural Resources (1979)

It might be a disappointment to advocates of consolidation that the plan for DNR did not include extensive immediate changes in the field of mapping,

charting, geodesy, and surveying as carefully worked out in the interdepartmental study of 1973. However, the Panel realizes that on each of the many occasions in the past when consolidation within this field has been recommended, the opposition from the involved agencies prevented any change. Also, in spite of the extensive study of the problems involved, there has been no clear agreement on how the various activities should be put together. The OMB in 1979 recommended a two-step strategy for reorganization of mapping, charting, geodesy, and surveying functions at the federal level: first, to bring the major agencies together in a new department and, second, at a later time to consolidate the surveying and mapping activities of the various components of the new department (Office of Management and Budget, 1979). Despite the problems involved, this Panel does not agree with this strategy. We stipulate that the civilian mapping, charting, geodesy, surveying, and multipurpose cadastre activities should, at this time, be consolidated in an independent agency. This agency can be placed within a new Department of Natural Resources, or as an interim measure within an existing department, but it should be independent of other organizational elements whose primary interests lie in other directions.

3.3 PROBLEMS DUE TO PRESENT ORGANIZATIONAL STRUCTURE AND PROPOSED REMEDIES

In the following sections we identify some of the major problems of the involved agencies as now organized and make suggestions for interim improvements. Brief descriptions of the present organizational structure of each agency are given in Appendix A.

A problem that is common to all surveying and mapping agencies is the difficulty in generating support for programs when the operating agency is deeply embedded in the "parent" organization. It is also a formidable task to prepare and have approved statements on national needs and goals for mapping, charting, geodesy, surveying, and the multipurpose cadastre when these activities are a minor function of the organization. This Panel believes that this problem, in the interim prior to the formation of a Federal Surveying and Mapping Administration, can be partially overcome by restructuring OMB Circular No. A-16 (see Section 3.3.7).

3.3.1 Geological Survey

The Geological Survey was one of several principal participants in the FMTF report study; and subsequent to the time of the report and its recommendations, the agency was able to implement several of the recommendations in coordination with, and with the cooperation of, many of the other federal mapping, charting, geodesy, and surveying (MCG&S) agencies. Notable

examples were the evolution of the National Mapping Program, which provided for more program flexibility resulting in more diverse products in response to user needs; the establishment of the National Cartographic Information Center; the development of intermediate-scale map series for the Soil Conservation Service for regional planning and the Bureau of Land Management for land management (this series was then made a part of the Geological Survey's standard quadrangle mapping program); the enhancement of an active and efficient program to produce orthophoto products; the evolvement of a digital data production program; and, most recently, the implementation of an internal reorganization to centralize the MCG&S activities of the Geological Survey in the new National Mapping Division.

Because of the decentralization of federal MCG&S activities, the focus has been turned to interagency coordination. The Geological Survey has had, and continues to schedule, periodic program and technical coordination meetings with federal agencies at all levels, depending on the particular subject matter and detail involved. These meetings with the Defense Mapping Agency, National Ocean Survey, Soil Conservation Service, Forest Service, Bureau of Land Management, Bureau of Indian Affairs, National Park Service, Tennessee Valley Authority, and others have all resulted in useful exchanges of information. There has not been any apparent indication that these past coordination efforts have been anything less than productive and contributory; but there is still much to be gained, both in financial and manpower resources, by the continuing effort for greater coordination.

The division of resources has been, and will continue to be, a deterrent to the progress of developing a single multipurpose digital cartographic data base. Each agency annually encounters difficulty in supporting and securing appropriations to support their particular digital-data-production and data-application activities. The questions that arise within the Office of Management and Budget as well as Congressional appropriation committees about the proliferation of digital-data-production activities have resulted in several studies as to which agency should be responsible for the digital cartographic data base. There was a strong recommendation in the FMTF report to develop a multipurpose digital cartographic data base, and this has been a primary effort of the Geological Survey for the past 6 years. However, while some funding was made available to proceed with hardware procurement and software development, funds to proceed with any level of significant data production have not been forthcoming (see Section 2.2.6 for recent funding proposal). The procurement, installation, and application of much of this new technology has also proven difficult to implement. Some of this can be attributed to the time needed to develop in-house expertise in technical areas, but it is impossible to quantify the loss (primarily in time) that has been caused by the division of the combined federal resources. There have been significant

contributions made to the Geological Survey by other agencies to advance the effort to implement a digital-data-production capability, but it cannot be projected where the program would be today if the federal efforts had been pooled. Therefore, there is a need for all agencies involved in digital data activities, be it production or data requirements to support programs, to communicate their needs and activities to the Geological Survey (see Section 2.2.6 for proposed lead-agency role).

The prevailing division of responsibilities for geodetic control, cadastral surveys, and map compilation continues to present a formidable problem. This was addressed in the FMTF report, and it remains a problem to the Geological Survey in undertaking the mission to produce multipurpose quality products. The availability of easily accessible and accurate control and public-land-survey data is a mandatory requirement in the production of mapping products, be they graphics or digital cartographic data. This problem is compounded by the time required initially to research the availability of data and the subsequent laborious effort to depict control and dissimilarly referenced cadastral information on cartographic products. The past and current efforts by the Geological Survey to compile much needed mapping of Alaska is a prime example of how a division of resources can affect efficient production of required cartographic information. There continues to be an ever-present need to consolidate the activities of cadastral surveys and the geodetic control network to focus upon one accurate and coordinated geographic reference system. The lack of sufficient funding is again the most resounding cause for the deficiencies in adequate geographic reference data; and it appears that the solution of this problem will remain elusive as long as there is a division of resources, both financial and manpower, among the federal agencies responsible for cadastral surveys, geodetic control surveys, and mapping. Consequently, there is yet a strong requirement that these important MCG&S activities be consolidated and coordinated to the fullest to ensure that adequate and accurate control and reference-system information are available when needed.

3.3.2 National Ocean Survey

The FMTF made recommendations for the expansion or extension of several of the programs or subprograms of the National Ocean Survey (NOS). Most of these recommendations implied the use of additional resources, that is, personnel, increased appropriations, or the transfer of funds from other agencies. The National Geodetic Survey of NOS has had difficulty in its attempt to respond to these recommendations.

The Task Force recommended "expansion of the National Geodetic Data Base to include selective control of less than second-order accuracy." The

NGS reports that the Geodetic Survey Data Base has been expanded to include third-order surveys from those agencies cooperating in the Federal Geodetic Control Committee. The Geological Survey and the Bureau of Land Management are converting survey data according to "Input Formats and Specifications–National Geodetic Survey Data Base." These data, plus all additional third-order data submitted in format, will be adjusted and included in the data base. Defense Mapping Agency data files have been furnished but are not in format and require analysis. The NGS lacks the resources to digitize, analyze, and adjust data that are not submitted in proper format. An increase in funding to review, analyze, and adjust data for the vertical datum program was requested but was denied. Currently, no vertical control data of less than second-order accuracy can be added to the data base. One must conclude that much of what has been accomplished was done by "reprogramming," that is, using funds that had been intended to support other parts of the total program.

The progress that the NGS has made in establishing the National Geodetic Survey Data Base is directly related to programs for the new adjustments of the horizontal and vertical control networks. The support for the data base–funds and personnel–has been drawn from the network programs or the total program. Neither procedure is a satisfactory solution. It is essential that additional funds be provided to support the National Geodetic Survey Data Base. Its functions will continue long after the new adjustments have been completed. The service to the total surveying community is a part of the responsibility assigned by OMB through the Federal Geodetic Control Committee.

The Task Force made similar recommendations involving the "continuance of cooperative (geodetic) programs" and the "building of a nationwide horizontal control system [with] resources . . . transferred from funds (estimated $1.5 million annually) which are expended by HUD clients for control surveys of unknown quality and reusability." The transfer of funds suggested in the latter recommendation has not taken place and would require formal action by the OMB and approval by Congress. The NGS has been able to make some progress through cooperative programs with a few states and other federal agencies, but the NGS has experienced a major difficulty because of personnel ceiling policies. A possible solution is the use of contractual services for more of the technical work. We believe that this can be done. It has been demonstrated by the Geological Survey for some of its cooperative mapping programs.

The recommendation that "the National Geodetic Program manager be given responsibility to (1) coordinate all federal civil Earth-model development activities that contribute to the National Geodetic Networks, and (2) accomplish geodetic measurements–to support polar-motion and crustal

movement investigations or studies" does produce some interaction between agencies that are investigating geophysical phenomena closely associated with these geodetic measurements. The precise repeat surveys made by the Geological Survey in California support their earthquake hazards and earthquake prediction programs. NASA is coordinating a broad national and international program in the application of space technology for the study of crustal dynamics and earthquake research. The NGS supports both of these agencies in these special studies. However, we believe that the fundamental practical reason for monitoring earth rotation and polar motion is to provide the corrections required for reducing astronomic and geodetic measurements to a common reference frame. The time-varying aspects that are the geodetic byproducts for the support of geodynamics frequently have a greater dramatic appeal to those responsible for providing the funding for the basic geodetic programs. These are matters to be resolved by interagency coordination with OMB and Congressional interaction. Reorganization or regrouping of agencies would not necessarily provide an effective solution.

Inquiries to the Division and District Offices of the Corps of Engineers disclosed that those offices expend more funds than does the NOS on hydrographic surveying. We recognize that the Corps of Engineers surveys are project oriented. However, many of those surveys are of the "single-purpose" classification. One of the statements in the 1973 report refers to another marine aspect: "The marine geophysics and oceanographic programs operated by NOAA, Coast Guard, and Navy, and those funded by the National Science Foundation are not sufficiently coordinated. Classified data collected by the Navy often are unavailable (even sanitized) to the civilian agencies." Greater coordination could be effected if the scope of OMB Circular No. A-16 were broadened to include hydrography and oceanography with direct control of ships, equipment, and personnel.

A review of the activities of the Federal Geodetic Control Committee (FGCC) indicates that the major efforts are in the interests of engineering geodesy with little consideration to the program for cadastral surveying, to the needs of research and development, and to the geodetic requirements for geophysics. *The FGCC should be restructured to include subcommittees for cadastral surveying, for research and development, and for geodynamics (including polar motion and crustal movement).*

3.3.3 Bureau of Land Management

The 1973 FMTF report contained two recommendations related to the cadastral survey program: (1) that the cadastral surveying functions of the BLM (excluding corner search and boundary marking) be transferred to a new central surveying and mapping agency and (2) implementation of a combined

geodetic/cadastral program in Alaska to meet requirements for state selection boundaries and for other cadastral surveys in Alaska and to provide geodetic control for multipurpose use. The report was critical of BLM's failure to locate monuments with reference to a geodetic network.

Today much of BLM's efforts in Alaska are coordinated with other surveying and mapping organizations, if possible, to provide multiple use of many of the monuments that are set. However, different priorities and funding arrangements among the surveying and mapping agencies prevent even closer coordination of these surveys. The BLM has improved its surveying methods and instruments since 1973 and now requires that public-land-survey monuments have geographic coordinate values (see Section 2.4.2).

The BLM Cadastral Survey Division is a service unit that is responsible for the legal cadastral surveys required by federal agencies. These surveys, which are extensive, consist of the initial surveys of federal public lands, including the Outer Continental Shelf lands, the resurvey of public lands, and boundary surveys required by various government agencies for the management of federal land. Because of deterioration of ancient surveys, resurveys of approximately 50 million acres are required. There is a large backlog of boundary survey requests that are awaiting execution; these are for the BLM's own resource-management purposes and for other federal agencies.

There is considerable concern about the effectiveness of the present cadastral survey program as evidenced by recent studies of the House of Representatives Appropriations Investigative Staff and by the BLM. If the present rate of surveying is continued, it will take several centuries to complete the initial surveys and necessary resurveys of the public lands. The BLM has not performed its delegated responsibility to service the needs of other federal agencies in a timely and effective manner. Several federal agencies have developed their own boundary-surveying competence because the BLM has not satisfied their requirements. With the increasing emphasis on multipurpose cadastres to provide timely information for the management of federal lands, the demands for boundary surveys will increase.

The organizational structure of the cadastral survey program within the BLM is the cause of some of the problems. The cadastral survey function is mixed with resource-management functions at several levels within the BLM organization. As indicated by its title, the Bureau of Land Management is primarily responsible for the management of lands, and its officials are oriented in that direction; this results in a low priority for cadastral surveys, as reflected in its budget, staffing, and policies. The management of the cadastral surveying program is fragmented, as noted in Appendix A.3, with a Division of Cadastral Survey in Washington, D.C., 12 state offices, and a Service Center with 2 branch offices, all of them reporting through several layers of management to the Director of the BLM. There is no national directive for the

cadastral surveying program; program operations are directed in each of several offices with no line responsibility nor authority for the operation of the program.

A reorganization of the cadastral survey program, either in a new surveying and mapping organization or at a higher level within the Department of the Interior, should be effected. The reorganization should provide a clear line of responsibility with the appropriate authority from the headquarters office to the field organizations. The cadastral survey organization must have more control over its program operations in order to establish policies, survey standards, practices, and priorities to serve the BLM and other federal agencies properly.

An increase in the number of highly qualified surveyors is needed to handle the changing demands of the cadastral survey organization from one that performs the initial surveys of public land to one that also provides the required resurveys and boundary surveys for the management of federal lands. The increasing number of administrative surveys marking the boundary between existing monuments, many of them intermingled with privately owned lands, requires professionals with the education and experience to handle complicated boundary problems. The development of multipurpose cadastre systems with the associated boundary surveying (including legal platting) and mapping requirements calls for surveyors who are educated and trained for that purpose. The position classifications and staffing levels of the cadastral survey organization must be increased so that the organization can hire and retain competent personnel to handle the surveying programs.

3.3.4 National Aeronautics and Space Administration

As discussed in Section 2.2.9 and Appendix A.4, NASA works closely with the other federal agencies and with the international community in the research and development of space techniques for mapping, charting, and geodesy applications. NASA has not encountered any serious problems due to federal organizational structures during its management of the National Geodectic Satellite Program once the question of security classification was resolved by the Congress in August 1962.

Some minor problems have developed over the years in Landsat and other satellite remote-sensing research and development that could be attributed somewhat to government organizational structure. Although NASA, from its conception, was designated a research organization, it found itself constantly being forced to respond to operational land-use and crop-inventory mapping and charting requirements. Yet none of the other federal agencies was willing to accept operation responsibility for a satellite remote-sensing land-observation system.

In November 1979, Presidential Directive/NSC-54 assigned NOAA responsibility for civil operational land remote-sensing activities. The initial system is to be based on Landsat technology, but NOAA is now responsible for defining and coordinating the future operational system technical requirements and for overall organization and management of the operational program. Once the NOAA operational program is fully implemented, NASA should be able once again to concentrate its efforts and resources on the research and development aspects of the program.

3.3.5 Other Agencies

The Federal Mapping Task Force identified 34 other federal agencies engaged in various aspects of domestic surveying and mapping efforts beyond the 5 principal agencies (4 civilian agencies and DMA). These agencies spent $162 million in fiscal year 1972. An additional 11 agencies were identified with $142 million in other activities such as oceanographic surveys, soil delineation, and geological investigations, which were considered closely enough related to be considered by the Task Force.

This Panel did not attempt a new inventory of these "other agencies" but believes that the volume of business in "other agencies" has significantly increased as the Task Force predicted. Increasing national concern about natural resources and the environment has added to the general increase in requirements for surveying and mapping data support.

Some improvement in the coordination between these widely proliferated surveying and mapping and related activities, and with those of the five principal domestic agencies, appears to have resulted from the Task Force efforts. There has, however, been no attempt within the Office of Management and Budget regularly to bring together information on all of these programs or to provide the overview necessary to assure the most effective and efficient use of this sizable portion of federal surveying and mapping resources.

In almost all of these agencies, the organizational submerging of these activities and the resulting limited visibility impede technical improvement and resource development essential to meeting the increasing requirements.

Many of the activities carried out in these agencies would, of course, have to be continued to provide local support services. Examples are local day-to-day cartographic services in the Bureau of the Census and the Tennessee Valley Authority (TVA), continuing local survey support to engineering projects in the TVA and the Mississippi River Commission, and local property surveys by Forest Service Offices and Corps of Engineers Districts. However, the major cartographic and survey projects of such agencies should be made a part of a single national program. The local programs should be carefully coordinated with the national program. Resources related to major projects

should be transferred to the proposed single civilian agency that would also have the coordination responsibility.

3.3.6 Rapport with Users

In addition to the problems that exist within and between agencies, this Panel believes that the relationship with the user community can be strengthened. The requirements for providing the essential services and products within government and to the private sector have been emphasized in earlier sections of this report. The reactions and responses of the users of these services must also be considered.

An effective means of communication between the agency and the user is the forum provided by national engineering and scientific societies. The American Congress on Surveying and Mapping, the American Society of Photogrammetry, the American Society of Civil Engineers, through its Division on Surveying and Mapping, and the American Geophysical Union, through its Section of Geodesy, have memberships drawn from government, private industry, and academia. Through the meetings and publications of these societies, there are opportunities to review existing programs, describe proposed programs, and present summaries of relevant research. These societies should be assigned a greater role in the development of national standards for surveying and mapping and the establishment of specifications for the procedures associated with these services. In order to use this broader base of expertise, the federal agencies should invite these societies to designate official representation on those governmental committees that involve the user sector.

3.3.7 Conclusion

In Section 2.3 we identified OMB Circular No. A-16 (see Appendix C) on coordination of surveying and mapping activities. In this section (3.3) it is evident that many of the problems within the present organizations result from the ineffectiveness of this circular and its use as a coordinating mechanism by the agencies. The essential reason is that the circular does not provide the authority for coordination. We believe that by restructuring Circular No. A-16 there can be significant improvements. Therefore, *we recommend that, even before the Federal Surveying and Mapping Administration is formed, the Office of Management and Budget revise the scope of Circular No. A-16 and add cadastral surveying; hydrography and oceanography (including ships, equipment, and personnel); and research and development for mapping, charting, geodesy, surveying, and the multipurpose cadastre. The revision should provide the authority for more effective coordination of federal civilian programs and should give recognition to the user community and to liaison with national engineering and scientific societies.*

4
Relationship of the Defense Mapping Agency to Civilian Agencies

4.1 BACKGROUND

Military and civilian mapping, charting, and geodesy programs are closely related and, though one serves national security and the other national development and their priorities and emphases differ, their products and services are similar and often identical. The relationships between federal military and civilian mapping, charting, and geodesy (MC&G) agencies are documented in legislation and in interdepartmental and interagency agreements. Since the establishment of the Defense Mapping Agency (DMA) in 1972, interagency contacts and cooperation between DMA and the civilian agencies have been simplified and appear to be operating more easily and effectively.

4.2 DEFENSE MAPPING AGENCY EXPERIENCE

Before the formation of the DMA in 1972 the mapping, charting, and geodetic groups within the Department of Defense (DOD) exhibited many of the problems now evident in the federal civilian agencies. The MC&G resources were spread among the military departments; facilities and capabilities were unnecessarily duplicated; the coordination of programs and technology was hampered by organization separation and lack of clear direction; serious deficiencies existed in meeting defense requirements for MC&G products; and resources for MC&G activities were being reduced within the military department budgets. Various management techniques, short of consolidation into a single agency, were tried to reverse these trends, but none was completely effective.

MC&G products and services were freely exchanged between the military services. For example, Air Force aeronautical charts were often in greater demand by the Army and aerial surveys by the Air Force and the Navy. Geodetic surveys by the Army were used by all three services. This resulted in each military department controlling resource allocations and production policy for items of critical importance to another service. To resolve this problem the Photographic and Survey Branch of the Office of the Joint Chiefs of Staff was established, but this Branch did not have sufficient management authority. The functions of this Branch were transferred to the Defense Intelligence Agency, in 1961 at the time of its establishment, and more specific responsibilities, including validation of requirements, product specification control, and program management, were added. However, a 1970 DOD study found that this arrangement was unsatisfactory since it did not provide control and direction over personnel and financial resources. The study recommended formation of a single military mapping organization. Further study led to the 1971 decision to establish the Defense Mapping Agency (DMA). This agency consolidated 80 percent of the DOD MC&G resources (8500 personnel and $305 million in fiscal year 1980). Those elements not included in DMA—mainly the hydrographic survey elements assigned to the Navy and the topographic troop units assigned to tactical commanders—are responsive to the program management of the DMA Director.

Over the 4 years prior to the formation of the DMA, MC&G personnel and budget were reduced by more than one fourth, even though the department resource managers had been advised of serious deficiencies in meeting national defense requirements. These out-of-proportion reductions were due to the MC&G programs being organizationally submerged in each of the military departments, where it did not receive significant visibility.

Within its first 3 years, the DMA achieved $36 million in savings through consolidations, standardization of procedures, and other management and technological improvements and reinvested these savings to increase productivity 18 percent. Improvements continue to be made. In addition, the DMA now presents its overall resource requirements for national defense directly to the Office of the Secretary of Defense, the Office of Management and Budget, and the Congress. The result has been that the more-critical requirements have received budgetary support. Consolidation has resulted in needs being handled in order of priority, in greater quantity, and in faster response time to users.

The DMA is able to focus timely attention on emerging MC&G requirements; today, 60 percent of its resources are used to produce digital and other nongraphical products for trainer/simulators and weapon-systems guidance as compared with approximately 35 percent at the time the DMA was formed. The flexible reprogramming of resources, procurement of common equipment and software for the production centers, and standardization of

the work force have enabled the DMA to respond to the rapid growth in requirements for digital products. Further benefits resulted from DMA's ability to establish digital data standards to which the users must conform, thus avoiding the cost of data-base proliferation.

4.3 GENERAL RELATIONSHIPS

In general, the DMA has responsibility for providing the maps, charts, and related materials, including geodetic, geophysical, and terrain data, required by the armed forces and other national security organizations. For areas of the United States and its territories, the DMA largely depends on products and services of the civilian MC&G agencies, producing coverage in these areas only when civilian-agency production cannot be responsive.

Since many foreign areas of national security interest are not accessible to DMA personnel for conventional surveying and mapping operations, the DMA supports an aggressive advanced technology program. Technology and data from this program of use to the civilian agency programs are carefully reviewed and transferred under procedures initiated by the Office of Management and Budget some 15 years ago. These include periodic technical exchange meetings and demonstrations for key officials of all federal MC&G agencies.

The DMA maintains working relationships and data exchange with many foreign government MC&G agencies. Since production problems faced by many of these agencies parallel those of civilian agencies, the DMA depends on the civil agencies to provide a part of the technical training furnished under these agreements.

4.4 CURRENT INTERACTIONS

The DMA interacts with the federal civilian agencies on topographic mapping, hydrographic and aeronautical charting, geodesy and geophysics, data libraries, digital data files and formats, research and development, and mobilization.

Requirements for military mapping of the territorial area of the United States, at scales of 1:24,000, 1:50,000, and 1:250,000, are included in the programs and budgets of the Geological Survey (GS). Special military maps are occasionally produced by the GS, sometimes with the assistance of the DMA. Those maps of the United States produced by the DMA are distributed through the GS.

The DMA, National Oceanic and Atmospheric Administration/National Ocean Survey (NOAA/NOS), and the Federal Aviation Administration, through the Inter-Agency Air Cartographic Committee, develop specifications

for joint-use charts and air information publications. For most charts and air information publications, separate military and civilian versions are required to fit the different operating environment. The DMA modifies the NOS civil edition reproducibles for its printings of U.S. area charts and air information publications, thus keeping DMA's activities in these areas to a minimum. NOS distributes any unique DMA air-chart publications.

NOAA/NOS produces the standard hydrographic charts for U.S. waters, and the DMA produces charts for the oceans; by statute, the DMA is responsible for the distribution of charts for non-U.S. waters to civilian maritime users. The DMA (Hydrographic/Topographic Center) and NOAA (NOS) represent the United States in the International Hydrographic Bureau, which fosters international standardization and exchange of sea navigation data. The DOD/Navy, DMA, and NOAA/NOS have formal coordination procedures, and the Oceanographer of the Navy maintains a full-time staff officer in the office of the NOAA Administrator. The NOS and the DMA operate separate public chart-distribution systems but collaborate on procedures and pricing.

The DMA depends on the geodetic and geophysical data (principally gravity and magnetic data) available from NOS programs for U.S. areas. The DMA carries out limited geodetic surveys in the United States, which are coordinated through the Federal Geodetic Control Committee, principally in connection with ICBM sites or weapons and system development, test, and calibration. The DMA has developed and operated systems for worldwide collection of data, has organized the data into libraries, and transforms the data into reference systems to meet weapon-systems needs. The DMA data applicable to U.S. areas are continuously reviewed for security classification and released to libraries operated by civilian agencies.

NASA, NOAA, and the DMA participate in the Satellite Geodesy Applications Board, developing and operating national geodetic satellite programs. Federal civilian agencies as well as the public utilize the Navy Navigation Satellite System (NAVSAT) operated by the DOD and are expected to have access to the DOD Global Positioning System for civilian geodetic positioning when it becomes available in the late-1980's.

The DMA maintains worldwide MC&G data libraries. Over the past 15 years, major parts of the U.S. data library have been discontinued or transferred to federal civilian agencies. The geodetic control and gravimetric data files of the United States have been transferred to NOS, and the development, by the GS, of the National Cartographic Information Center has been encouraged. The DMA maintains the foreign geographic names file and provides staff support to the U.S. Board on Geographic Names for foreign areas. The U.S. geographic names file is maintained by the GS.

The digital terrain data for U.S. areas, initiated by the DMA, have been transferred to the GS, where they are maintained for joint civilian-military

use. The rapid growth in civilian and military applications for digital cartographic data as well as the increased use of computer systems in photogrammetric and cartographic processes is generating large digital data files. The GS, NOS, and DMA are working toward standardization in data file formats to simplify cross-utilization of data files and to ensure rapid response in emergencies.

In addition to the periodic technical exchange programs for key MC&G officials of the DMA and the civilian agencies noted in Section 4.3, the GS, NOS, and DMA have formal agreements for cooperation in research and development. These provide for regular briefings on program plans and status and encourage cooperation at each phase of any project. Joint funding and joint working groups are other features. The three agencies recently agreed to procurement of identical production systems unless serious operating problems make a specific system impractical for any of them. As the technical complexity of systems continues to grow, the ability to exchange data and products at all phases of the MC&G process can be adversely affected by system differences. The three agencies have also agreed to regular exchange assignment of technical staff to improve the transfer of ideas and information and to assure greater general awareness of the full scope of federal MC&G activities.

4.5 1973 REPORT RECOMMENDATIONS APPLICABLE TO THE DEFENSE MAPPING AGENCY

In general, the 1973 report recommendations would continue the basic relationships currently existing between the DMA and the civilian mapping community, except that, if there were a consolidated surveying and mapping administration, the multiple interactions described in Section 4.4 would reduce to a single interaction.

Some of the recommendations, particularly those applicable to release of classified gravity, bathymetric and similar data, and technology exchange in mapping, charting, and research and development, identified need for increased coordination. Considerable progress has been made since the 1973 report (see Section 2.2.3).

The conclusion of the 1973 report that independent civilian and military agencies are in the nation's best interest is still considered to be valid. A case can probably be made for increased operating efficiencies through consolidation in some production operations. Any cost savings identified, however, would be heavily outweighed by the importance of maintaining separate and positive control over program priorities to assure support for the two widely differing national requirements.

4.6 INTERACTION BETWEEN THE DEFENSE MAPPING AGENCY AND A SINGLE CIVILIAN AGENCY

The separation between military and civilian agency activities is well defined, and cooperation appears to be working well but is subject to extenuating circumstances at any given time. Some areas, for example, hydrographic surveying, charting, and chart distribution, probably need further review to determine areas where adjustments, such as joint civil–military operations, may be desirable. Other minor adjustments would undoubtedly result in additional efficiency. Obviously this can be more easily accomplished if the DMA could interact with a single civilian surveying and mapping agency.

The DMA is heavily dependent on the civilian mapping community for MC&G products and services to meet national security requirements for U.S. areas. The extent and quality of this support can be significantly improved by the increased emphasis, efficiency, and interactive effectiveness that can be expected from the establishment of a civilian Federal Surveying and Mapping Administration.

5
Conclusion

The basic sciences of surveying and geodesy provided the foundation for the engineering profession. Mapping and charting grew through the exploration and control of the lands and seas. All of these areas are now widely recognized as individual and separate sciences and professions.

Since surveying and mapping have always provided critical support to a wide variety of national programs, most countries have given their national programs significant visibility by establishing central agencies, such as that of Surveyor General, with wide responsibility. The United States has tended to permit proliferation of its national program through many agencies. Since this proliferation tends to reduce the effectiveness and efficiency of these critical support programs, there have been many attempts through U.S. history to bring greater order. For example, earlier in this century a Federal Board of Surveys and Maps was established, and for a time the Executive Office of the President had a Survey and Maps Coordinator in the Office of Management and Budget (OMB, then the Bureau of the Budget). This latter function faded away since it did not fit into the budget/program review structure that is the basis for current OMB organization and operations.

Without an overall program manager, there has been no overview of the total federal surveying and mapping program for many years, even though the scope of the program has greatly expanded, spurred on by increasing national needs for data on one hand and by surging scientific and technological progress in these areas on the other.

Without a central manager, the growth of surveying and mapping capabilities took place through the years in many federal agencies to provide the

specific survey data, maps, and charts they needed to support their own programs. A principal example is the growth of topographic mapping in the Geological Survey of the U.S. Department of the Interior. As other agencies learned of these products and recognized their value to their programs, arrangements were made to assign responsibility for production that would meet the needs of other agencies as well as those of the producing agency. Such coordination of programs has been useful. However, with the control of resources remaining subordinate to the interests of the principal functions of the agencies, it is unlikely that resources allocated to surveying and mapping would ever be consistent with the real national requirements for services and production.

To a large extent, federal surveying and mapping have been responsive to current particular project support requirements or to a particular product need identified by individual user agencies or by specific public or state or local government user groups. In many instances, the cost of work needed to meet such specific needs would not be greatly increased if all national requirements were accommodated in designing and carrying out the work. A significant waste of resources is suffered when the work done for one purpose must be repeated to meet other requirements. A central manager responsible for determining all national requirements and for developing technology, facilities, and programs to meet them should be able to eliminate this waste. Furthermore, all the national interests that depend on surveying and mapping services would realize gains in their effectiveness, efficiency, and economy from the improved availability of this critical data base.

The 1973 Task Force report noted that a major improvement has already been realized in U.S. military surveying and mapping as a result of the President's decision to eliminate organizational proliferation and centralize program direction by establishing the Defense Mapping Agency. This Panel is convinced that similar benefits would derive from the transfer of surveying and mapping and related functions of the several federal civilian agencies to a Federal Surveying and Mapping Administration.

Appendix A: Present Organizational Structure

A.1 U.S. GEOLOGICAL SURVEY

The U.S. Geological Survey (USGS) is organizationally structured into four programmatic Divisions: the National Mapping Division, Geologic Division, Water Resources Division, and Conservation Division; two program-support Divisions: the Administrative Division and Computer Center Division; and two programmatic offices under the Office of the Director: the Office of Natural Petroleum Reserve in Alaska and the Office of Earth Science Applications.

The National Mapping Division was organized during fiscal year 1980 to consolidate mapping, charting, geodesy, and surveying programs and activities of the Geological Survey from the former Topographic Division, the former Publications Division, and the former Geography Program (of the Office of Earth Science Applications). In support of the USGS mission to provide information about the earth and its physical resources, the National Mapping Division provides geographic and cartographic information, maps, and technical assistance and conducts related research responsive to national needs. To accomplish this mission, the Division collects, compiles, and analyzes information about natural and man-made features on the earth's surface and documents changes as appropriate; produces and maintains series of accurate and up-to-date general-purpose base maps and thematic maps such as land-use/land-cover maps; develops and maintains a digital geographic–cartographic data base for multipurpose needs and assists Survey divisions and other federal and state agencies in developing and applying spatial data;

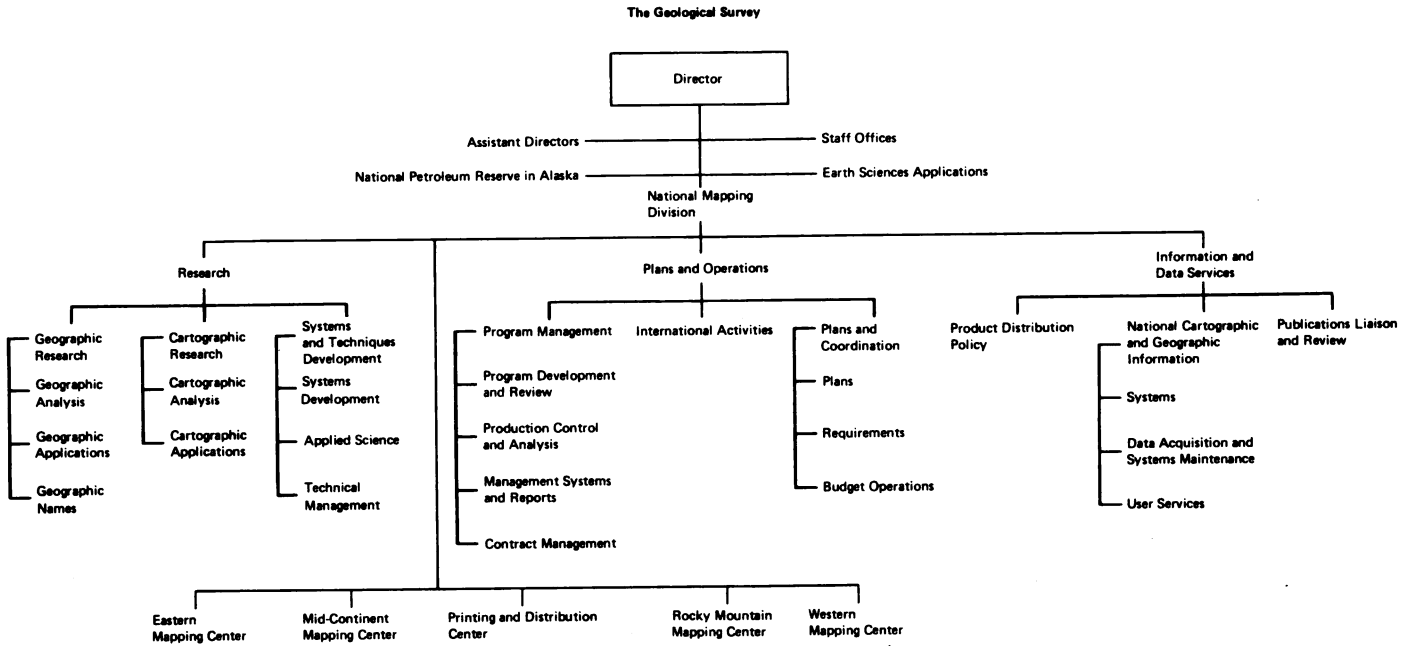

The Geological Survey
Director
Assistant Directors
Staff Offices
National Petroleum Reserve in Alaska
Earth Sciences Applications
National Mapping Division
Research
Geographic Research
Geographic Analysis
Geographic Applications
Geographic Names
Cartographic Research
Cartographic Analysis
Cartographic Applications
Systems and Techniques Development
Systems Development
Applied Science
Technical Management
Plans and Operations
Program Management
Program Development and Review
Production Control and Analysis
Management Systems and Reports
Contract Management
International Activities
Plans and Coordination
Plans
Requirements
Budget Operations
Information and Data Services
Product Distribution Policy
National Cartographic and Geographic Information
Systems
Data Acquisition and Systems Maintenance
User Services
Publications Liaison and Review
Eastern Mapping Center
Mid-Continent Mapping Center
Printing and Distribution Center
Rocky Mountain Mapping Center
Western Mapping Center

TABLE A.1 National Mapping Division Budget Figures for Fiscal Year 1980

Program Element	Positions	Funding ($ Thousands)
Primary quadrangle mapping and modernization	1092	38,341
Map revision and orthophotoquads	375	14,288
Digital mapping	32	2,627
Small-scale, intermediate-scale, and special mapping	185	14,453
Cartographic and geographic information and data services	26	3,058
TOTALS[a]	1710	72,767

[a]These figures do not include positions and funding associated with the former Publications Division.

conducts geographic, cartographic, and reproduction research utilizing modern technology and equipment; provides thematic mapping support to Survey divisions and other federal agencies; prints topographic, geologic, hydrologic, land-use, and other thematic maps; and operates information and technical assistance centers that gather, index, analyze, and catalog geographic and cartographic information. The Division also makes available maps, imagery, spatial data, and related information; provides assistance in selecting, acquiring, and using geographic and cartographic products; and designs, prints, and distributes maps of the National Atlas. It coordinates federal mapping activities and provides leadership in the development and advancement of surveying and mapping technology.

Table A.1 gives fiscal year 1980 budget figures for personnel and funding for the National Mapping Division.

A.2 NATIONAL OCEAN SURVEY

The National Oceanic and Atmospheric Administration (NOAA) is managed from a structure that comprises five main-line components including the Office of Oceanic and Atmospheric Services. The Office of National Geodetic Survey (NGS) is one of five basic functional offices of the National Ocean Survey (NOS) within the structure of Oceanic and Atmospheric Services.

The NOS program activities and responsibilities comprise mapping, charting, and surveying services and operations of the NOAA fleet and ship bases.

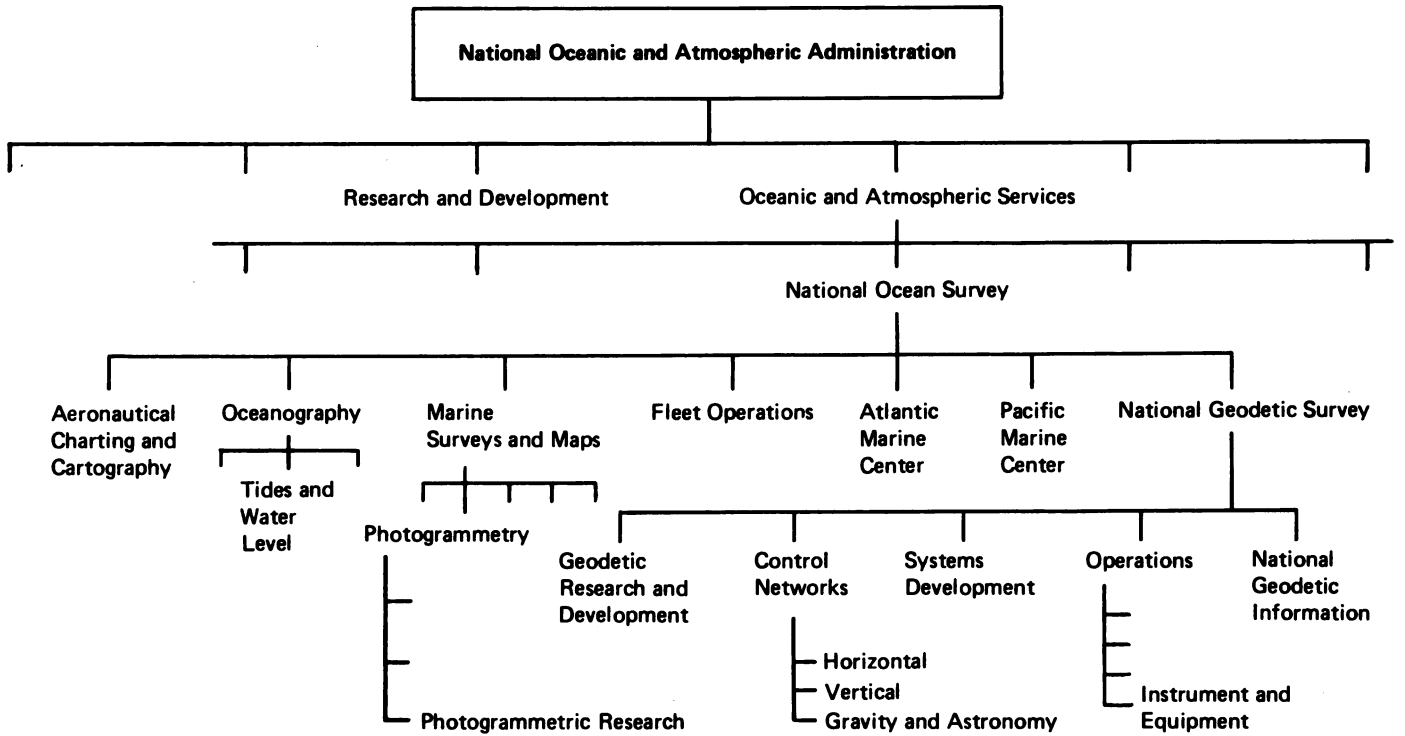

National Oceanic and Atmospheric Administration
Research and Development
Oceanic and Atmospheric Services
National Ocean Survey
Aeronautical Charting and Cartography
Oceanography
Tides and Water Level
Marine Surveys and Maps
Photogrammetry
Photogrammetric Research
Fleet Operations
Atlantic Marine Center
Pacific Marine Center
National Geodetic Survey
Geodetic Research and Development
Control Networks
Horizontal
Vertical
Gravity and Astronomy
Systems Development
Operations
Instrument and Equipment
National Geodetic Information

TABLE A.2 National Ocean Survey Budget Figures for Fiscal Year 1980

Office	Positions	Funding ($ Thousands)
Geodetic Surveys and Services	339	13,236
Nautical Charting Services	404	15,695
Ocean and Coastal Mapping	220	10,468
Aeronautical Chart Services	464	21,577
Fleet Operations[a]	887	42,951
TOTALS	2314	$103,927

[a]Includes support for all NOAA components; in particular, National Ocean Survey, Environmental Research Laboratories, National Fisheries Service, and National Weather Service.

These program activities are responsive to NOAA missions, which include providing for the following:

1. Charts and related information for safe navigation of marine and air commerce;
2. Ocean and coastal surveys and maps;
3. Marine resource assessment, monitoring, and prediction;
4. Marine ecosystem investigations, analysis, and ocean dumping;
5. Ocean and coastal management, conservation, and protection;
6. Geodetic surveys and services;
7. Public forecast and warning services; and
8. Supporting research, development, and engineering.

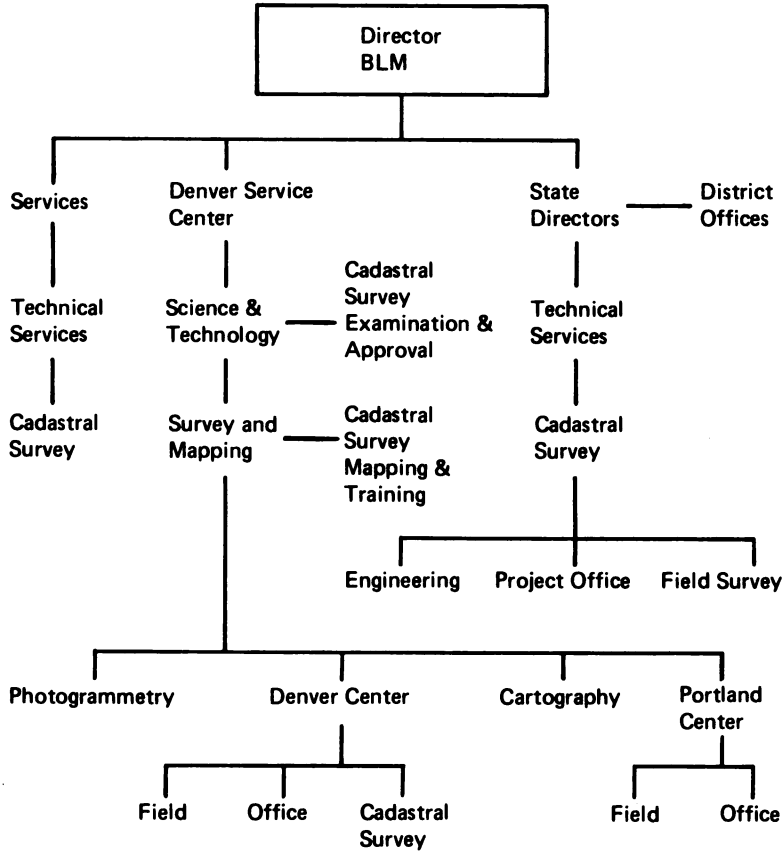
The NOS program response to the NOAA mission is funded and managed under (1) Geodetic Surveys and Services, (2) Nautical Charting Services, (3) Ocean and Coastal Mapping, (4) Aeronautical Charting Services, and (5) Fleet Operations, as shown in Table A.2.

Table A.2 gives the fiscal year 1980 budget figures for personnel and funding for the National Ocean Survey.

A.3 BUREAU OF LAND MANAGEMENT

The cadastral survey organization of the Bureau of Land Management (BLM) is responsible for the legal boundary surveys required by the federal government. There are three general classifications of surveys conducted by the BLM: (1) the initial surveys of the federal public lands to create boundaries that may be used for the management and sale of public lands, including the

Bureau of Land Management
Division of Cadastral Survey



Outer Continental Shelf (OCS); (2) the maintenance of the Public Land Survey System (PLSS), with resurveys and remonumentation of the original surveys; and (3) the boundary surveys required by the BLM and other federal agencies to delineate management tracts within the public lands or between federal and nonfederal lands (administrative surveys).

The Cadastral Survey organization is divided into three general groups:

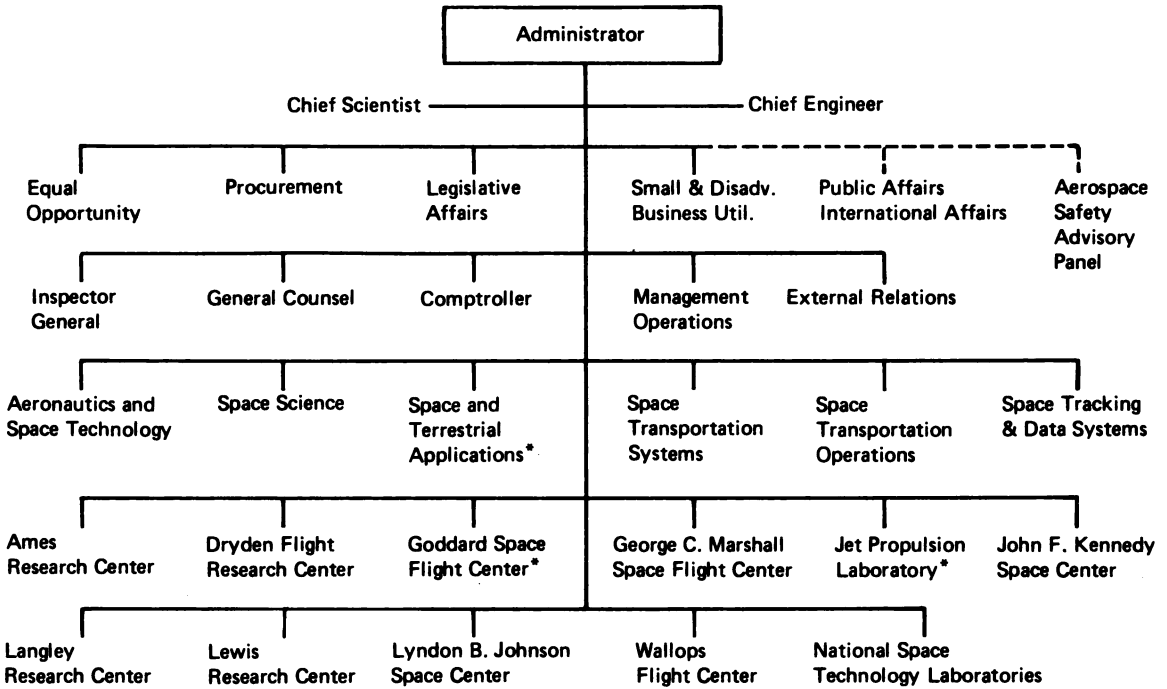
1. Division of Cadastral Survey in Washington, D.C., including the Division Chief and a small staff that reports through channels to the Director of BLM and is responsible for program development, technical development, public liaison, and technical guidance.
2. State offices (12) that report through channels to the State Director and are responsible for the cadastral surveys within their area, including original surveys, resurveys, and some reimbursable surveys for other agencies. Each state office develops its program, conducts the field surveys, and processes the information in accordance with prescribed standards and policies. The state office is usually responsible for the preparation, filing, and maintenance of the official cadastral records within its area.
3. Service Center Offices in Denver, Colorado, and Portland, Oregon, that report through channels to the Director of the Denver Service Center and are responsible primarily for the cadastral surveys required by other federal agencies. The program management responsibilities are similar to the state organizations. In addition to the surveys described above, the Denver Service Center reviews and approves the survey plats, prepares the OCS protraction diagrams, and conducts personnel training and development programs.

During 1980 the cadastral survey organization had 373 permanent positions (Division of Cadastral Survey, 12; State Offices, 253; and Service Center, 108), and expenditures totaled $17.8 million.

A.4 NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

As provided for in the National Space Act of 1958, The National Aeronautics and Space Administration (NASA) has the responsibility to conduct scientific research in space for the benefit of mankind. Studies of the earth from space and of the earth as a planet are within the purview of the responsibilities assigned by the Act. In fulfillment of these responsibilities and to support other efforts having need of geodetic information, such as satellite orbit computation, NASA has pursued an active research program in the development and application of space techniques for the accomplishment of geodetic measurements.

National Aeronautics and Space Administration



*The principal geodetic activities of NASA are conducted by groups in these organizations.

NASA is administered from a structure that comprises six program offices, which, supported by the various staff offices, manage the activities of eleven field installations. All geodetic, geodynamic, and earth-related geophysical research activities are managed through the Office of Space and Terrestrial Applications (OSTA). OSTA is supported by the Office of Space Tracking and Data Systems for data acquisition and initial processing, by the Office of Aeronautics and Space Technology for common Supporting Research and Technology, by the Office of Space Science for Spacelab-associated missions, and by the Office of Space Transportation Operations for Shuttle-associated missions.

In August 1962, in view of NASA's responsibilities for international cooperation in the field of space sciences, Congress directed the Agency to implement a National Geodetic Satellite Program to meet the needs of all interested parties, scientific and military. This program was initially coordinated by the Geodetic Satellite Policy Board chaired by NASA with members from the Department of Commerce/National Oceanic and Atmospheric Administration and the Department of Defense/Defense Mapping Agency. In 1978, it was reconstituted as the Satellite Geodesy Applications Board (SGAB) with the same mission and participants.

Within OSTA, the Resource Observation Division has primary responsibility for the conduct of the geodetic, geodynamic, and solid-earth geophysics programs. The three branches (Geodynamics, Renewable Resources, and Non-Renewable Resources) work closely in developing spaceborne and ground-based sensors for measuring and monitoring the earth's magnetic and geopotential fields, as well as small-scale motions (centimeters/year displacement and deformation) of the global tectonic plates. Coordination is also effected with the Oceanic Processes Branch of the Environmental Observations Division on research related to the earth geoid and geopotential model as affected by ocean circulation.

In addition to the SGAB coordination, NASA has initiated extensive interagency and international coordination of its space-related crustal dynamics and gravity field measurement programs.

Funding for geodetic-related space activities in fiscal year 1981 is approximately $90 million; the staff comprises approximately 50 civil servants.

A.5 DEFENSE MAPPING AGENCY

The Defense Mapping Agency (DMA) is responsible for providing mapping, charting, and geodetic (MC&G) support to the Department of Defense. In addition, the DMA has statutory responsibility for providing nautical charts and marine navigation data for the use of all vessels of the United States and of navigators generally.

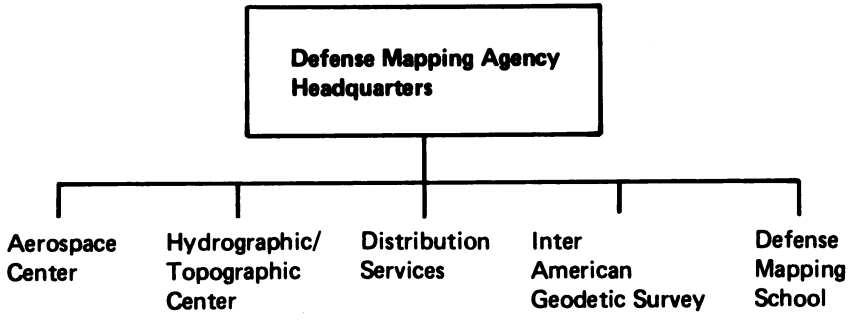
TABLE A.3 The Defense Mapping Agency Budget Figures for Fiscal Year 1980

Component	Positions	Funding ($ Thousands)
HQ DMA	189	$ 29,862[a]
Aerospace Center	3655	121,434
Hydrographic/Topographic Center	3828	128,566
Office of Distribution Services	452	14,630
Inter-American Geodetic Survey	163	7,303
Defense Mapping School	196	2,697
TOTAL	8483	$304,492

[a]Includes $20,012 in research and development funds to support all DMA activities.

The DMA was created in 1972, as the result of a Presidential decision, to achieve greater efficiency and effectiveness through consolidation of the MC&G elements then assigned to the Army, Navy, and Air Force. Hydrographic surveys remained with Navy and topographic troop units with the Army and the Marines, but these activities are under DMA program management.

The DMA is organized into a Headquarters and five Components: the Aerospace and Hydrographic/Topographic Centers carry out the production mission; the Office of Distribution Services stores and distributes all DMA maps and charts; the Inter-American Geodetic Survey provides technical support to cooperating Latin American countries as a means of meeting U.S. national security needs; and the Defense Mapping School trains military personnel in MC&G subjects. DMA personnel are assigned at a total of 59 locations worldwide. The fiscal year 1980 personnel and funding levels are given in Table A.3.



Appendix B: Prior Surveying and Mapping Studies

The most detailed study of federal surveying and mapping functions and operations prior to the 1973 Report of the Federal Mapping Task Force on Mapping, Charting, Geodesy and Surveying of the Office of Management and Budget was conducted by the Science Advisory Board of the National Research Council at the request of the Bureau of the Budget. The Board transmitted, on November 13, 1934, a "Report of the Committee on Mapping Services of the Federal Government together with Exhibits." Included in this report were summary statements of studies completed prior to 1934 (pages 170-179) and a summary of the report (pages 141-142). All of these reports are out of print and, as the conclusions reached in these reports are germane to the current proliferation in surveying and mapping, the summaries are reproduced on the following pages.

EXHIBIT A

PREVIOUS INVESTIGATIONS OF FEDERAL SURVEYING AND MAPPING ACTIVITIES

CONGRESSIONAL COMMITTEE, 1842

The question as to whether the hydrographic work of the Coast Survey should be under control of the Navy or of a civil department of the Government was early agitated. In 1842 this agitation led to a Congressional investigation which severely scrutinized the operations of the Coast Survey. The result was a strong endorsement, by the investigating committee, of the principles on which the Survey was being conducted under civil control.

PRESIDENT'S BOARD OF 1843

In 1843 Congress passed an act providing that a board of officers, to be organized by the President, should formulate a permanent plan for conducting coastal surveys. The Board, consisting of two naval officers, four Army officers, and three officers of the Coast Survey, unanimously recommended that the coastal surveys should remain, as they then were, under civil control. This recommendation has since governed surveys of the coastal waters of the United States, although not extended to surveys of foreign waters and of the high seas, nor always to coastal waters of territorial possessions of the United States.

AMERICAN ASSOCIATION FOR THE ADVANCEMENT OF SCIENCE, 1858

In 1858, following further efforts of the Navy Department to gain control of coastal surveys, the American Association for the Advancement of Science appointed a committee of twenty leading scientists to investigate the work of the Coast Survey. The report of this committee paid high tribute to the accomplishments of the Survey under civil control and recommended that the general plan of organization then in force be continued.

NATIONAL ACADEMY OF SCIENCES, 1878

Congress in 1878 directed the National Academy of Sciences to investigate the scientific surveys being conducted under jurisdiction of the War Department and the Interior Department and the surveys of the General Land Office; and to report to Congress a plan for surveying and mapping the

territories of the United States which would secure the best results at the lowest possible cost. The report of this investigation favored leaving surveys for strictly engineering purposes, such as the improvement of rivers and harbors and the irrigation of public lands, to the Corps of Engineers of the War Department; advised no change in respect to the Lake Survey (War Department) because "this is now nearly completed"; recommended that the various geological and geographical surveys then in operation under different departments be abolished and a single United States Geological Survey substituted therefor; and further recommended that the Coast and Geodetic Survey be transferred from the Treasury Department to the Department of the Interior, that its name be changed to the United States Coast and Interior Survey, and that in addition to its regular functions it assume direction of the topographical surveys of the public domain and the land-parceling surveys then under control of the General Land Office. Had the Academy realized that more than half a century later the Lake Survey would still be prosecuting its labors at a cost of from $150,000 to $200,000 annually, there can be little doubt that it would have recommended transfer of its activities to the proposed Coast and Interior Survey. The consolidations actually proposed were well conceived. The one respecting consolidation of all geological and geographical surveys in a single unit was carried into effect with excellent results. Had the proposal to concentrate hydrographic, topographic and land-parceling surveys under unified control been carried out, higher standards of work would have been achieved, the mapping program expedited and millions of dollars saved to American taxpayers. Opposition in Congress defeated this proposal.

Allison Joint Commission, 1884-86

From 1884 to 1886 a joint commission of Congress, known as the Allison Commission, investigated the Signal Service, Geological Survey, Coast and Geodetic Survey and the Hydrographic Office of the Navy Department, with a view to securing greater efficiency and economy in the administration of these bureaus. After securing the advice of the National Academy of Sciences and hearing a large volume of expert testimony from other sources, four of the six Commission members submitted a report which opposed transfer of the Coast and Geodetic Survey to the Navy Department on the grounds that the Coast and Geodetic Survey had been in successful operation over half a century "with men at the head of the Survey confessedly of the highest scientific attainments during all this period," that such transfer would impose on naval officers "duties entirely inconsistent with their profession" and require their permanent employment in what would be to them new scientific work, and that there was "no reason either on the score of

efficiency or economy for making the change." With respect to the Hydrographic Office, all six members of the Commission praised its administration, recommended that no change in the organization be made and that " this office be maintained by appropriations from year to year in its present state of efficiency." Two members of the Commission submitted a minority report recommending transfer of all hydrographic work of the Coast and Geodetic Survey to the Hydrographic Office of the Navy. This bill failed of passage, both houses of Congress concurring in the views of the majority report maintaining the status quo.

Dockery Joint Commission, 1893-95

From 1893 to 1895 another joint commission of Congress known as the Dockery Commission investigated the executive departments of the Government with a view to securing greater efficiency and economy in their administration. This Commission employed two expert accountants to make a report on " Public Surveys." The report of the experts, which was approved by the Commission and submitted to Congress, recommended that all land surveys, including geologic and scientific investigations, be consolidated in one bureau; and all water surveys in another. The bureau to direct the land surveys, to be known as the " United States Survey " and located in the Interior Department, .should combine the geodetic work of the Coast and Geodetic Survey, the topographic and other work of the Geological Survey and the land-parceling work of the General Land Office. Transfer of the hydrographic work of the Coast and Geodetic Survey to the Navy Department, and consolidation with the Hydrographic Office under the name " Hydrographic Bureau " were held to be the most effective means of unifying water surveys. The report necessarily suffers from the fact that it was prepared by experts in the field of public accountancy who possessed no special knowledge of the scientific fields they were directed to investigate. While well conceived in part, it included recommendations of very doubtful expediency based on incomplete knowledge of facts. A bill embodying their recommendations failed to receive the sanction of Congress.

Congressional Discussion, 1900

In 1899 and 1900 the activities of the Coast and Geodetic Survey and of the Hydrographic Office of the Navy were the subject of an interchange of letters between the Secretary of the Treasury and the Secretary of the Navy. The activities of the two bureaus were in the Spring of 1900 elaborately discussed before appropriation committees and on the floor of the House. Hearings were held, memoranda and other documents were submitted, including the correspondence cited above and debates extended over a period of

some weeks. It was, in effect, another Congressional investigation, in which alleged efforts of the Hydrographic Office to extend its field of operations and take over work of the Coast and Geodetic Survey were severely scrutinized but without overt action of consequence resulting.

CLEVELAND COMMISSION, 1910-12

From 1910 to 1912 a Commission appointed by President Taft under authority granted by Congress conducted an investigation into the executive departments of the Government for the purpose of recommending legislation designed to secure greater efficiency and economy. This body was known as the President's Commission on Economy and Efficiency, or the Cleveland Commission. An "Interior Department Committee" was appointed by the Secretary of the Interior to cooperate with the Cleveland Commission; and this Committee in its turn appointed a "Sub-Committee on Surveying, Map Making and Graphic Reproduction." The report of the Sub-Committee recommended, by a vote of 4 to 3, that the land-parceling or subdivisional surveys of the General Land Office and the topographic surveys of the public lands should be consolidated under a single administrative officer; and that the surveys of the Indian Office and of the Reclamation Service should be placed under the same officer, "so far as practicable and feasible." These recommendations were approved by the Interior Department and forwarded to President Taft. A new Interior Department Committee was then organized by the Secretary of the Interior to formulate a plan of consolidation along lines suggested in the Sub-Committee's report. This new Committee, by a vote of 3 to 1 (the representative of the General Land Office dissenting), recommended that "both subdivisional and topographic surveys be combined under the Director of the United States Geological Survey." No specific reference was made to the surveys of the Indian Office and the Reclamation Service. The Acting Secretary of the Interior, Samuel Adams, by direction of the President, sought legislation which would permit consolidating the subdivisional surveys of the General Land Office with topographic surveys of the Geological Survey; but the proposal apparently died in committee.

FEDERAL BOARD OF SURVEYS AND MAPS, 1920-22

In 1919 President Wilson created by Executive Order a Federal Board of Surveys and Maps, the purpose of which was "to coordinate the activities of the various map-making agencies of the executive departments of the Government, to standardize results, and to avoid unnecessary duplication of work," the Board being directed, among other things, "to make recommendations to the several departments or to the President for the purpose of coordinating all map-making and surveying activities of the Govern-

ment." The Board's powers appear to have been largely if not wholly advisory, and its members differed as to whether they were authorized to make recommendations involving *consolidation* rather than *coordination* of the surveying and mapping services. From 1920 to 1922 the Board, aided by a Committee on Coordination, wrestled with the difficult problem of improving the Federal surveying and mapping situation. A first report, making tentative suggestions for limited consolidation, was by the Board referred back to the Committee for further consideration. A second report, rejecting the principle of consolidation, was similarly referred back to the Committee for further study of the problem. Several additional proposals in turn met with the same action. Finally, on the morning of October 11, 1921, the Chairman of the Board reported that the Joint Committee on Reorganization of Executive Departments (see below) would probably recommend some consolidation of mapping and surveying activities, that a report embodying recommendations by the Federal Board of Surveys and Maps would be welcome, but that it would have to be submitted immediately.

Thereupon the Committee on Coordination presented at that same meeting a report (in effect a revision of its first report), recommending consolidation of primary surveying and mapping activities in a proposed new "Bureau of Surveys and Maps." To get quick action the rules of procedure requiring that a report shall be sent to all member organizations at least two weeks prior to final action were waived without objection. After long discussion it was voted, 9 to 7, "to consider only a reallocation of the five primary surveying and mapping agencies to existing departments or new departments which may be created, with a view to letting the Secretaries of those Departments call upon the Board of Surveys and Maps for suggestions as to methods of consolidation." That same afternoon the Board, by a vote of 10 to 6, reconsidered the morning action, and after variously amending the report of the Committee on Coordination recommended, by a vote of 10 to 2 with 6 members absent and not voting, far-reaching consolidation in the following terms:

"The Committee, therefore, recommends the creation of a Bureau of Surveys and Maps in some existing or new Department. This Bureau should comprise six major branches, or divisions, and should perform the primary surveying and mapping functions of existing Federal agencies, as follows:

1. *Geodetic Branch*
   - *a.* Geodetic work of Coast and Geodetic Survey.
   - *b.* Geodetic work of Geological Survey.
   - *c.* Geodetic work of Lake Survey.
   - *d.* Geodetic work of Hydrographic Office, U. S. Navy.
   - *e.* Control work of General Land Office.
   - *f.* Magnetic work of Coast and Geodetic Survey.

2. *Topographic Branch*
   - *a.* Topographic mapping of Geological Survey.
   - *b.* Topographic mapping of War Department not directly connected with military operations.
   - *c.* Base map work of Bureau of Soils insofar as practicable.
   - *d.* Topographic work of Forest Service insofar as practicable.
   - *e.* Topographic work of Bureau of Indian Affairs insofar as practicable.
   - *f.* Topographic work of Reclamation Service insofar as practicable.
   - *g.* Topographic work of Coast and Geodetic Survey.
3. *Hydrographic Branch*
   - *a.* Hydrographic work of Coast and Geodetic Survey.
   - *b.* Hydrographic work of Hydrographic Office, U. S. Navy.
   - *c.* Hydrographic work of Lake Survey.
   - *d.* Tidal and current work of Coast and Geodetic Survey.
4. *Cadastral Branch*
   - *a.* Surveying work of General Land Office.
   - *b.* Cadastral work of Forest Service.
   - *c.* Cadastral work of War Department.
5. *Cartographic Branch*
   - *a.* State and International Map Compilation work of Geological Survey.
   - *b.* State and United States Map Compilation work of General Land Office.
   - *c.* Work of Map Information Office.
6. *Map Reproduction Branch*
   - *a.* Reproduction work of Geological Survey.
   - *b.* Reproduction work of Coast Survey.
   - *c.* Reproduction work of Lake Survey.
   - *d.* Reproduction work of War Department, not directly connected with military operations.
   - *e.* Reproduction work of Hydrographic Office, U. S. Navy."

It was provided that the consolidations recommended should take place gradually, and with due regard to the importance of not unduly disturbing existing operations.

Soon after this report was submitted to President Harding very strong dissatisfaction developed with respect to the hasty manner in which final action had been secured. Vigorous efforts within the Board to get the report recalled from the President's hands were ultimately defeated by a vote of 11 to 6, in part apparently because of the view repeatedly expressed, that to take such action would embarrass the President and place the Board in an impossible position. The proposals of the Board were not included in the reorganization scheme later submitted by President Harding and his Cabinet (*see* below), nor in the report made to Congress by the Joint Committee on Reorganization of the Executive Departments (*see* below).

The stormy history of the report of the Federal Board of Surveys and Maps, the fact that its provisions were finally revised and adopted under pressure and with undue haste, and the further fact that the Board does not

appear to have been in possession of all the facts necessary for a critical judgment on many of the difficult problems involved, inevitably weaken the force of its specific recommendations concerning individual agencies. The report has real value, however, as evidence that within the Federal services themselves there then existed a strong sentiment that some form of consolidation was essential to efficiency and economy in mapping activities.

Some twelve years later the Science Advisory Board addressed to the Federal Board of Surveys and Maps an inquiry for information on possible plans for consolidation, which inquiry was carefully considered by the Executive Committee of the latter organization and replied to by the Chairman, in part in the following terms: "Although the Board in October, 1921, passed a resolution on this matter, a decided difference in opinion existed. The Board probably would not have taken a definite stand except for great pressure to take immediate action. Later an attempt to withdraw the report was made. Defeat of this effort was largely due to a feeling that the Board having once submitted a report to the President would be in an embarrassed position if it later withdrew it." After noting that substantial changes in Board membership had occurred since 1921, and that a proposal for widespread consolidation would probably be decidedly opposed by the present membership, the communication stated: " It is the firm opinion of the Executive Committee of the Board of Surveys and Maps that no consolidation of the mapping services of the United States Government could be effected that would improve the existing situation, either as to economy or efficiency."

President Harding and His Cabinet, 1920-23

Congress in 1920 created the Joint Committee on Reorganization of the Executive Departments (*see* below), and a few months later requested the President to participate actively in the study to be made. President Harding, after long delay " caused solely by the difficulty which has been encountered in reconciling the views of the various persons charged with the responsibility of administering the executive branch of the government," submitted to the Committee proposals prepared by the President and his Cabinet. These proposals did not relate specifically to surveying and mapping activities, but did include transfer of non-military activities of the War Department, including the Board of Engineers for Rivers and Harbors and the Mississippi River Commission, to the Department of the Interior; transfer of marine activities of the War Department, including the Lake Survey, to the Department of Commerce; transfer of the Hydrographic Office from the Navy Department to the Department of Commerce; transfer of the Bureau of Public Roads from the Department of Agriculture to the Department of the Interior;

transfer of functions of the Federal Power Commission, an independent establishment, to the Department of the Interior. During hearings on the President's plan the Secretaries of the Interior, of Commerce, and of the Navy advocated transfer of the geodetic work of the Coast and Geodetic Survey to the Geological Survey, but without success.

BROWN JOINT COMMMITTEE, 1920-24

From 1920 to 1924 a Joint Committee on Reorganization of the Executive Departments, created by Congress and headed by Walter F. Brown, a personal representative of the President, investigated the administrative services of the Government to determine what redistribution of activities should be made in the interest of efficiency and economy. This Committee gave careful study to the proposal of President Harding and his Cabinet that the Hydrographic Office be transferred from the Navy Department to the Department of Commerce in order to consolidate charting activities in one department but concluded there was no duplication of work by these departments and that there was real economy in having naval vessels make soundings in foreign waters and on the high seas. The consolidation of engineering and public works enterprises in a single department was likewise opposed. The Joint Committee, however, in a majority report signed by four members recommended transfer of the Lake Survey from the War Department to the Department of Commerce, and transfer of the Bureau of Public Roads from the Department of Agriculture to the Department of the Interior. A minority report signed by two members opposed this last transfer. A bill embodying the recommendations of the majority of the Joint Commission, although publicly supported in the Republican national platform of 1924, by the former Secretary of Commerce under President Wilson and by President Coolidge in a message to Congress, failed to gain adequate support in that body.

SECRETARIES HOOVER AND WORK, 1927

In 1927 Secretaries Hoover and Work continued their efforts to have the geodetic work of the Coast and Geodetic Survey transferred to the Geological Survey, and had introduced into both houses of Congress a bill designed to effect the change. Although this proposal originated with Secretary Hoover, in whose Department the Coast and Geodetic Survey was located, and was welcomed by Secretary Work whose Department included the Geological Survey, their combined efforts were fruitless. Vigorous opposition to the change was voiced in scientific circles of the nation and within the Government, and the bill designed to effect the change failed to make headway in Congress.

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PRESIDENT HOOVER, 1932

On December 9, 1932, President Hoover, in order to reduce expenditures and increase efficiency in the government service, transmitted to Congress the results of a study of executive departments in which he had long been interested. These results were embodied in a series of ten or more Executive Orders which made wholesale regroupings or consolidations of 58 executive agencies. By these Orders many non-military activities of the Corps of Engineers, including those of the Mississippi River Commission and the Board of Engineers for Rivers and Harbors, were transferred from the War Department to a new Division of Public Works in the Department of the Interior, which was also to include the Bureau of Reclamation and the Geological Survey already in the Department, and the Bureau of Public Roads which was transferred there from the Department of Agriculture; the Bureau of Indian Affairs and the National Park Service of the Interior Department were grouped with other services in a new Division of Education, Health and Recreation in that Department; the Forest Service and the Bureau of Chemistry and Soils in the Department of Agriculture and the General Land Office in the Department of the Interior were transferred to a new Division of Land Utilization in the Department of Agriculture; the Hydrographic Office of the Navy Department and the Lake Survey of the War Department were transferred to the Coast and Geodetic Survey, the consolidated group forming part of a new Merchant Marine Division in the Department of Commerce. As some of the above named agencies are primarily surveying and mapping agencies, while the others engage in important surveying and mapping activities, it will be seen that the Executive Orders issued by President Hoover were expected to effect extensive regrouping and some consolidation of surveying and mapping agencies. The Orders were accompanied by a statement prepared by the Bureau of the Budget explaining the reasons on which the proposed transfers were predicated. On January 19, 1933, all of the Orders were rendered null and void by resolution of the House of Representatives.

AMERICAN ENGINEERING COUNCIL, 1933

In March, 1933, the American Engineering Council, which had long been interested in efforts to improve the surveying and mapping situation, issued a "Report on Reduction in the Expenditures of the Federal Government." The report advocated immediate consolidation by Executive Order of Federal surveying and mapping activities, referred to the plan of consolidation proposed by the Federal Board of Surveys and Maps in 1921 (*see* above) and said: "Council thoroughly approves the plan and recommends that it be put into effect at the earliest possible date." The Council's approval of the

consolidation plan of the Board of Surveys and Maps was not based on detailed studies conducted by the Council itself, but rather upon the personal experience and resulting convictions of engineers composing the organization and its affiliated bodies. The endorsement of a particular program of wholesale consolidations, hastily adopted by an organization the Executive Committee of which later pronounced against any consolidation, cannot, therefore, carry weight as respects the specific transfers advocated. But it does have value as accurately reflecting the widespread conviction in engineering circles that extensive consolidation of surveying and mapping activities would promote efficiency and economy within the government service.

Committee on U. S. Coast and Geodetic Survey, 1933

On July 31, 1933, the Chairman of the Business Advisory and Planning Council of the Department of Commerce appointed a Committee on the United States Coast and Geodetic Survey. This Committee in its "Report on the First Meeting Held August 21, 1933" (only three weeks after its appointment and, therefore, without lapse of time for adequate study of surveying and mapping problems) endorsed the October, 1921 proposals of the Federal Board of Surveys and Maps favoring wholesale consolidation of surveying and mapping activities. Its endorsement must be weighed as that of a committee studying and reporting on a single bureau, without sufficient opportunity to investigate conditions in the many other bureaus affected by proposals of the Board of Surveys and Maps. Thus regarded, the report of the Committee is valuable for its approval of the work of the Coast and Geodetic Survey, evidenced by its recommendation that this unit be made the nucleus of the proposed Bureau of Surveys and Maps; and as further evidence of the widespread demand for some form of consolidation of mapping activities. It cannot have weight as regards particular transfers recommended in the Board's 1921 proposals.

SUMMARY OF REPORT

This report is based on six months' intensive study of 28 Federal agencies engaged in surveying and mapping, made at the request of the Science Advisory Board by a committee of seven specialists selected for their qualifications in the fields primarily concerned. *All statements, conclusions and recommendations are based on unanimous agreement of the Committee.*

Federal surveying and mapping activities are found to be enormous in volume, of confusing variety and to have developed independently and often without correlation in many different executive agencies. As an inevitable result duplication of effort and waste of funds have sometimes resulted. Subsidiary surveying services have occasionally grown to great mapping units in abnormal situations, causing inefficiency and waste.

Remedies are deemed imperative. But the Committee has only attempted to suggest what is reasonable and practicable. That many Federal agencies must have subsidiary surveying services under their control is fully recognized. *Out of the 28 agencies studied, three and parts of one other are selected for immediate consolidation* (Coast and Geodetic Survey, Lake Survey, International (Canada) Boundary Commission, Topographic Branch and Division of Engraving and Printing of the Geological Survey). This will form a nucleus toward which certain other Federal surveying and mapping activities will naturally gravitate in time, or to which they can be added if and when deemed expedient.

The consolidation proposed will unite under efficient expert direction *all major Federal land and water surveys* except cadastral surveys of the General Land Office. All changes contemplated, except that involving the International Boundary Commission, have repeatedly been urged following previous studies of this problem. No change advocated will seriously affect any existing executive agency. In most cases the need of reform has long been felt either by administrative heads of the agencies affected; or by the scientific and technical personnel, or by both.

Immediate savings aggregating *several hundred thousand dollars annually* are anticipated. If a large program of mapping the country is inaugurated, involving the expenditure of fifteen million dollars each year for a specified period, direct economies due to consolidation will aggregate *between two and three million dollars or more annually,* while indirect savings to the people arising out of possession of the map may be measured in terms of *many millions of dollars.*

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The proposed consolidation will provide the most efficient instrument possible for utilizing large numbers of men in all parts of the country in a well-considered program for pushing rapidly to completion the standard topographic map of the United States. Since the completion of this map will save the nation many millions of dollars annually thereafter, a large mapping program is in effect a relief measure in which every dollar spent will soon pay large dividends, and continue to pay them far into the future. Under such conditions *relief becomes a sound investment in monetary as well as in human values.*

A suggested form of Executive Order indicates how the Committee's recommendations can be effected.

Subsequent to 1934, a number of reports pertaining to surveying and mapping were issued. Although not so extensive as the 1934 and 1973 reports, these later reports are listed here for completeness.

Task Force Report on Public Works (Appendix Q); prepared for The Commission on Organization of the Executive Branch of the Government (Hoover Commission) (January 13, 1949).

A Bill to Provide an Accelerated Program for Surveying and Mapping of the United States, in Territories and Possessions, and for Other Purposes, H.R. 6900, Hearings before the Committee on Public Lands, House of Representatives (February 3, June 3, 9, and 14, 1950).

Survey of the Federal Domestic Cartographic Activities, Requirements and Organization, Bureau of the Budget Staff Report (June 30, 1962).

Review of Geodetic Surveying Activities within the Federal Government, prepared by the Comptroller General of the United States (GAO Report B-133188) (January 25, 1967).

Memoranda for the President of the United States, *Establishment of a Department of Natural Resources, Organization for Social and Economic Programs*, submitted by The President's Advisory Council on Executive Organization (February 5, 1971).

Appendix C: OMB Circular No. A-16, Revised May 6, 1967

The Bureau of the Budget (now the Office of Management and Budget) issued Circular No. A-16 in 1953. This Circular described the responsibilities of federal agencies involved in surveying and mapping activities. Appended to this Circular were Exhibits, occasionally revised, that dealt with such items as classification and standards of accuracy of geodetic control surveys and procedures for programming and coordination of federal topographic mapping activities. In 1967, Circular No. A-16 was revised, delegating responsibility for aspects of surveying and mapping to various federal departments. The contents of the latter Circular are given on the following pages.

EXECUTIVE OFFICE OF THE PRESIDENT
BUREAU OF THE BUDGET
WASHINGTON, D.C. 20503

May 6, 1967 CIRCULAR NO. A-16
Revised

TO THE HEADS OF EXECUTIVE DEPARTMENTS AND ESTABLISHMENTS

SUBJECT: Coordination of surveying and mapping activities

1. *Purpose.* This revised Circular describes the responsibilities of Federal agencies with respect to coordination of the Federal surveying and mapping activities described in paragraph 2, below. It rescinds and replaces Circular No. A-16, dated January 16, 1953. Exhibits A, B, C, and D to that Circular will remain in effect until replaced pursuant to paragraph 4 of this Circular.

2. *Coverage.* The coordinating procedures established by this Circular extend to all surveying and mapping activities financed in whole or in part by Federal funds which:

a. Can contribute to the National Topographic Map Series of the United States and outlying areas of sovereignty and jurisdiction, the National Atlas of the United States of America, the National Networks of Geodetic Control, or such other national geodetic control and topographic mapping programs as may be established; or

b. Result in cartographic representation of international boundaries other than those of the United States with Canada or Mexico.

Surveying and mapping activities conducted or supported by a Federal agency to meet specific program needs of the agency which are not met by the national surveying and mapping programs specified in paragraph 2a above and which cannot practicably or economically contribute to the national programs are excluded from coverage. Determination of the surveying and mapping activities which are required to meet program needs is the responsibility of the program agency. However, evaluation of the potential contribution of those activities to a national surveying or mapping program should be made by such agency in consultation, or pursuant to a general agreement, with the responsible agency as described below.

3. Responsibility for coordination.

a. *The Department of the Interior* is responsible for the National Topographic Map Series of the United States and outlying areas of sovereignty and

jurisdiction and for the National Atlas of the United States of America. It also operates the Map Information Office, which collects and furnishes to potential users information concerning aerial photography, topographic mapping, and survey data available from Federal agencies for general use, and the sources from which they may be obtained.

In carrying out these functions the Department exercises Government-wide leadership in assuring coordinated planning and execution of its national topographic mapping, National Atlas, and map information activities and the cartographic activities of other Federal agencies related thereto, including activities financed in whole or in part by such agencies, to the end that:

(1) The topographic mapping, National Atlas, and map information needs of Government agencies and the public at large are met in the most expeditious and economical manner possible with available resources;

(2) All mapping activities financed in whole or in part by Federal funds contribute to the national topographic mapping program when it is practicable and economical to do so; and

(3) Aerial photography, topographic mapping, and survey data produced by Federal agencies are conveniently accessible for use in meeting the cartographic needs of other Federal agencies and federally assisted programs.

The Department will also arrange, through periodic meetings or other appropriate means, for an exchange of information among Federal agencies concerning technological developments in civilian agencies with respect to cartographic activities.

b. *The Department of Commerce* is responsible for the National Networks of Geodetic Control and publishes status maps of geodetic control which meet the standards for inclusion in the national networks.

In carrying out this function the Department exercises Government-wide leadership in assuring coordinated planning and execution of its national geodetic control surveys and the related survey activities of Federal agencies, including activities financed in whole or in part by such agencies, to the end that:

(1) The geodetic control needs of Government agencies and the public at large are met in the most expeditious and economical manner possible with available resources; and

(2) All surveying activities financed in whole or in part by Federal funds contribute to the National Networks of Geodetic Control when it is practicable and economical to do so.

c. *The Department of State* exercises Government-wide leadership to assure that cartographic representations of international boundaries, other than those of the United States with Canada or Mexico, by all Federal agencies are consistent and conform to United States foreign policy.

4. *Establishment of coordinating mechanisms.* Each agency named in paragraph 3 above will, in consultation with other Federal agencies concerned, establish such standards, procedures, interagency agreements, and other mechanisms as are necessary to carry out its Government-wide coordinating responsibilities and to replace, where required, Exhibit A, B, C, or D of this Circular.

5. *Responsibilities of other Federal agencies.* Each Federal agency is responsible for (a) cooperating as requested in the development of appropriate coordinating mechanisms; (b) supplying necessary information to the coordinating agency concerning its cartographic requirements, programs, activities, and products; and (c) conducting its surveying, mapping, and product distribution activities in a manner which provides effective Government-wide coordination and efficient service to the general public.

6. *Differences among agencies.* Any major differences which cannot be resolved through consultation among agencies with respect to the coordination of cartographic activities covered by this Circular may be referred by the head of any agency concerned to the Director of the Bureau of the Budget.

PHILLIP S. HUGHES
Acting Director

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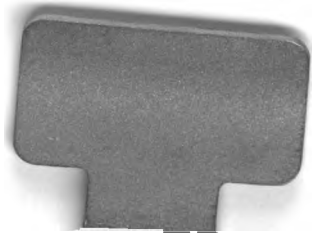
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